

T660 Programming Guide

This guide explains how to program a T660 Digital Delay Generator using its command line interfaces, and acts as a reference for its many commands.

In this guide, introductions and explanations are provided for syntax- and command-specific behaviors of the T660. However, for the documentation of most command behaviors, it is assumed the reader is familiar with the relevant features of the T660 and their operation. This guide is not intended to explain how features work or interact, although at times unintuitive interactions will be noted to avoid potential confusion. For descriptions or overviews of how features work, see the T660 manual.

If the reader is new to the T660, even having previously used a T560 or T564, it is recommended to read the overview section to better understand the T660's feature-rich programming capabilities.

Programming Interfaces

Command Sessions

The T660 is programmed and interfaced via one or more *command sessions*, also referred to as *sessions*. A session is a bidirectional ASCII data stream where a client sends the T660 *commands* and the T660 replies with *responses*.

Commands may be run across different command sessions simultaneously. While the T660 can only execute one command at a time (excluding the wait commands — these commands do not block other command sessions during their execution), it will attempt to interleave commands originating from multiple command sessions on a per-command basis, allowing for near seamless, but still atomic, multiprocessing operation.

This allows the T660 to be controlled from multiple sources simultaneously, allowing, for example, reading the unit's live status while it is being primarily controlled by another computer.

The behavior of command sessions is identical regardless of the physical or digital interface used to create or access them, and their behavior is consistent across all interfaces. This includes basic text formatting (newline and whitespace compatibility is identical across all interfaces), and command interpretation (commands are interpreted identically regardless of the session's interface of origin), for example.

Session Modes

Introduction

The T660 provides two fully functional, feature-complete command sets to program the T660 with. These two *session modes* provide style compatibility with other Highland Technology products:

- P500-style (compatible with the P400 and P500 Digital Delay Generators)
- T560-style (compatible with the T560 and T564 Digital Delay Generators)

The distinction between these two session types is in their interpretation of command lines and responses. *P500-style sessions* parse command lines the same way the P500 and P400 do and send responses expected by existing P500 infrastructure, and *T560-style sessions* interpret commands and send responses the same way the T560 and T564 do.

Each session mode has its own set of commands, syntax, and parsing rules independent from the other. The only assured similarity between these two modes is how they interpret command delimiters: both modes delimit commands on the same line with the ; (semicolon) character, and command lines with a line delimiter (<cr>, <cr><lf>, or <lf> for command lines, and <cr><lf> for response lines). Both modes also offer substantial command additions compared to the T560 and P500 Digital Delay Generators.

Although the T660 is a successor and is intended to be as backward compatible as possible with the T560, the T560 style of command programming is not recommended for use in new designs. Because the P500 style of programming offers more extensible and thorough programming, future T660 features may not be added to the T560-style command set despite being accessible through P500-style commands. Despite this, existing T560-style features will not be removed, so existing applications built for the T560 should require no modification to interface with a T660.

T560 Compatibility

The T560-style mode has been designed to be as accurate as possible to the T560 and T564, replicating command responses nearly character-for-character and behavior nearly identically. Existing infrastructure designed for T560 interfaces should require minimal changes, if any, to work completely with the T660.

P500 Compatibility

P500-style parsing and commands, although extremely representative of and accurate to the P500's commands, include a few changes to existing commands to facilitate their use in the T660. The T660 cannot fully replace a P500's functionality, so some features such as channel voltage control are missing as commands.

Since the T660 implements various new features compared to the P500, such as additional gate modes, commands have been expanded where appropriate in a backwards-compatible way. For example, in the `GATE:MODE <mode>` command, where the P500 used 0 through 4 to represent possible gate connector configurations, the T660 uses 0 through 10.

There are a couple notable differences where existing commands have been modified to reduce confusion:

- The `TRIGger:SOURce <source>` command uses `INT` to represent the internal 80 MHz trigger source, instead of the frequency synthesizer, which is now represented by `SYN`.
- The `SYSTem:COMMunicate:SOCKeT:NetStat?` command additionally replies with DHCP status.

Behaviorally, there are also a few differences:

- `TRIGger[:EXTeRnal]:PREDiv <divisor>` applies to all trigger sources, not just external ones. The `EXTeRnal` keyword has been made optional to reflect this.
- Changing trigger source no longer implicitly executes a `STOP` command.
- Burst functionality behaves slightly differently in edge cases, but is generally fully compatible and so uses the same commands.

For timing configuration, the T660 only supports the P500's FAST insertion delay mode.

Switching Modes

At any time, session mode may be changed using the session mode commands. These commands are provided in both T560-style and P500-style command sessions. The session mode commands have been designed such that in either P500-style or T560-style mode, the text `ZM P500` and `ZM T560` sets the session mode, and `ZM?` queries the current mode (the response is the same regardless of active session mode).

Although command session type may be changed mid-session, there are some restrictions on switching, and how these command sessions are initiated and accessed. See [Session Interfaces](#) for details regarding these restrictions.

Using simultaneous P500-style and T560-style modes across multiple sessions is fully supported. Note, though, that executing any channel timing-related command (including reading statuses) will change channels to delay-width mode across all P500 sessions. See P500-style Channel Timing Mode for details.

Session Interfaces

A single T660 may be interfaced through any combination of physical ports and session interfaces simultaneously. Active command sessions may be initiated over any physical port or session interface and are all treated with equal priority. The number of command sessions initiated over Ethernet is unlimited, so care should be taken to ensure that the number of active Ethernet connections remains reasonable.

There are four session interfaces that accept or service command sessions, a comparison of which is provided below:

Session Interface	Physical Port	Max. Connections	Default Session	Session Types
Serial	Serial	1	T560-style	All
TCP (Port 2000)	Ethernet	Unlimited	T560-style	T560-style only
TCP (Port 2001)	Ethernet	Unlimited	P500-style	P500-style only
Isochronous ¹	Embedded	1	T560-style	All

¹ This interface is only available to T660-10 units.

The behavior of command sessions is otherwise identical regardless of the physical or digital interface used to create or access them, and their behavior is consistent across all interfaces. This includes basic text formatting (newline and whitespace compatibility is identical across all interfaces) and command interpretation (commands are interpreted identically regardless of the session's interface of origin), for

example. For compatibility, a single exception to this rule applies to the isochronous interface; see [Isochronous Interface](#) for details.

The connection details for each of these interfaces are explained in greater detail in the following sections.

Serial Interface

The serial interface can be accessed via the RS-232 serial port. Upon power-up, the serial interface operates using a T560-style session. The default baud rate is 38400 baud, however this may be changed using commands.

The RS-232 port uses the following connection parameters:

Baud Rate	9600 19200 38400 (default) 57600 115200 230400 460800
Data Bits	8
Stop Bits	1
Parity	None

TCP Interfaces

There are two TCP socket interfaces, one for each session mode, both of which are serviced via a network connection using the T660's built-in Ethernet port. Both interfaces are serviced with a socket server hosted by the T660 on their respective ports: port 2000 is used for creating T560-style sessions, and 2001 for P500-style sessions.

All socket interfaces are expected to be connected to via a network client with a raw connection. Although Telnet connections will typically work, they will invalidate the first command line and produce an error once the first line delimiter is sent.

Isochronous Interface

The Isochronous Interface is intended to be used for embedded applications, and only exists for the T660-10. It operates identically to the [Serial Interface](#) but with a different physical layer protocol. This interface is accessed using the embedded connector headers of the T660-10.

There is a single exception to cross-interface identical behavior that only applies to the isochronous interface: the FEod command also stops the frames engine. See FEod for details.

Details regarding the physical protocol and operation this interface are available upon request as part of the *T660 Manual -10 Addendum*.

Errors and Error Codes

Command Errors

The T660 does not use the same error codes as either the T560 or P500. In addition to this, the T660 can provide more detailed error messages if desired. Whenever an error is encountered while executing a command (or query), the error code as well as the brief and full descriptions may be provided as the command's response. This behavior is configurable on a per-session basis.

When an error occurs as a result of any command in the current session, diagnostic information about the error is sent as a response instead of the command's normal response. This response will always include at least one question mark character (?), and may include:

1. A second question mark character (?)
2. The error code, as a two digit decimal integer
3. The error name
4. A brief description of the error

Error Verbosity

The T660 introduces a concept referred to as *error verbosity*. In both T560-style and P500-style sessions, the response given when a command encounters an error can be configured with the `VERbose ERrors <level>` and `SESSion:VERBosity[:ERRor] <level>` commands, respectively. T560-style sessions default to an error verbosity of 0, while P500-style sessions default to an error verbosity of 3 at the start of the session. Every time a session's mode is changed, its error verbosity is changed to the session mode's default level.

Error verbosity is represented as an integer, with lower values representing less information. A table of verbosity levels is provided below:

Level	Information	Example
0	The text ??	??
1	Error code	?32
2	Error code and name	?32: Unexpected argument
3	Error code, name, and brief description	?32: Unexpected argument: Too many arguments provided for the specified command.

List of Error Codes

Even though the T660 can print error descriptions when errors occur, a reference of all error codes and their descriptions is useful, and is provided below.

Code	Brief Description	Explanation
Behavioral		
?03	Line abort	Received an abort character while processing the command line.
?04	Not implemented	The requested operation is not yet implemented.
?05	Command protected	The requested command is not allowed to be executed due to its protected status.
?06	Feature required	Operation not permitted because a required feature is not enabled on this device.
?07	Illegal operation	The current state of the device does not permit the requested operation.

Code	Brief Description	Explanation
?08	Settings conflict	A change of configuration could not be applied because the current configuration is incompatible with the change.
Internal		
?10	Unexpected error	Unexpected error while performing the operation, device state may be invalid.
?11	Software error	Unexpected software error.
?12	Hardware error	Unexpected hardware error.
?13	Calibration error	Missing, unsupported, or invalid calibration.
Command		
?20	Syntax error	Invalid command syntax.
?21	Command name expected	Expected a command or query, but did not find one.
?22	Command not recognized	The named command was not recognized as a command or query.
?23	Subcommand expected	The named command is only recognized as a subsystem and cannot be executed.
?24	Command is query-only	Command was specified as a command, but it only exists as a query.
?25	Command is set-only	Command was specified as a query, but it only exists as a command.
Argument		
?30	Invalid argument format	Argument could not be interpreted as the expected data type.
?31	Argument expected	Not enough arguments provided for the specified command.
?32	Unexpected argument	Too many arguments provided for the specified command.
?33	Invalid argument value	Argument is out of range or is an illegal value.

T560-Style Syntax

Recognized Characters

In T560-style sessions, only the following characters are parsed from command lines. All other characters are ignored.

Recognized characters include:

- Alphabetic: A-Z and a-z (all alphabetic characters are case insensitive)
- Numeric: 0-9
- Special characters: ., ,, +, -, and /
- Whitespace: <space>, <tab> (both are treated as a space)
- Command delimiters: ; and :
- Command line abort: <ETX>, <BS>, <ESC>,

All other characters, such as * and ?, are ignored. The command line abort characters clear the currently buffered command line; once received, the current command line is discarded and not executed, and the next character is treated as the start of the command line.

Note that compared to the T560, the T660 additionally parses the +, -, and , characters.

Basic Syntax

Each *command* consists of one or more command keywords, optionally followed by an alphanumeric argument. Multiple commands may be sent in a single *command line*, separated by semicolons (or colons). When a full line is received, indicated by a line delimiter (<cr>, <lf>, or <cr><lf>), the buffered line's commands are executed in the order received.

Each command generates a *response*, and each command line generates a *response line*, consisting of all the commands' responses in the order executed, separated by semicolon (;) characters. Execution of the next command in a command line will block until the current command finishes writing its response via the session it was run in. This means that commands run over a network session can execute much faster than those over the serial session.

Every response line is terminated with the carriage return and line-feed characters (<cr><lf>). Some commands produce multi-line responses, such as the S*Tatus* command. In these cases, these rules still apply, and these commands' responses are treated as though they do not include a line delimiter.

For every semicolon that is present on a command line, the response line generated will include the same number of semicolons. This means that an empty command, consisting only of whitespace, will still produce a semicolon in the response line. Whitespace is not preserved between valid commands, and is ignored. For example, ; ;FI; FI;;; responds with ;;OK;OK;;;, and TL; might respond with 1.250;.

A "blank" input line, consisting only of whitespace, evokes the response T660<cr><lf>.

Command Structure

Command keywords may be fully spelled out or sent as their first two letters; only the first two letters are significant. In this documentation, a word that has two possible forms is written with the short form capitalized, and the rest of the word in lower-case letters.

All commands and arguments are case insensitive. Whitespace (spaces and/or tabs) is required to separate keywords from arguments, or keywords from other keywords.

For example, TR*igger* indicates that the short form is TR, and the long form is TRIGGER, both of which are recognized as the same command. A command line consisting of TR RE; FI would respond OK;OK, and perform both commands in sequence.

Query commands are requests for specific data. A query is often a "set"- type command without an argument.

Time-set commands are expressed as channel delays and widths, with the four pulse outputs identified as A, B, C, or D, corresponding to the four channel output connectors.

Each received command will evoke a reply indicating the execution status of the command. For query commands, the reply is the requested data. For all other commands, successful completion will yield a reply of OK, and an error will produce a ?? response and terminate the line.

If an error occurs while processing a command, the reply ?? will be returned. If multiple commands are present on a command line, and any command produces an error, the erroneous command will respond with the error and no further commands will be executed on the same line. This means that an error will always terminate a command line and be followed by a line delimiter (<cr><lf>).

Configuring Pulse Outputs

Unlike P500-style command sessions, T560-style commands do not provide a method of both configuring and applying timing, channel, and trains (*Frames and Trains* required) configuration in a single command; all commands only affect the pending configuration.

To apply configured settings, an install must be performed, either manually (via the `INStall` command) or with the autoinstall function.

Autoinstall

Autoinstall is a configurable feature that is saved as part of the global configuration (it may be set at power-on to a user-configured value). The autoinstall configuration is global to the device and applies to all current and future T560 sessions immediately when changed.

Each session contains its own, independent autoinstall flag. Whenever the pending configuration is modified, the autoinstall flag is set. Whenever an `INStall` or `QQueue` command is run, the flag is cleared.

An autoinstall occurs at the end of a command line whenever the autoinstall flag is set. This issues either an `INStall` or `QQueue` (*Frames and Trains* required) command, after which the flag is cleared. No additional response will be generated as a consequence of the autoinstall.

Autoinstall has three modes:

Mode	Description
0	Autoinstall is disabled.
1	At the end of a command line where the pending configuration has been modified, an <code>INStall</code> command will be issued.
2	At the end of a command line where the pending configuration has been modified, a <code>QQueue</code> command will be issued. This is only available to models with the <i>Frames and Trains</i> feature.

Autoinstall mode may be changed via the `AUTOinstall <mode>` command.

Reading Data

Queries

Various value-setting commands may be sent without an argument, in which case they become queries of the associated value.

For example, ADeLay evokes the reply 02.123456789123, which represents the delay setting for the rising edge of the channel A output, in seconds.

Numerical replies to queries will be in fixed-point decimal numeric form, with embedded commas included if [verbose mode](#) is enabled.

Status Responses

Most configuration and status information is not available to query directly and must be parsed out of single- or multi-line status responses. These responses can be queried via status commands which are typically the name the feature to be queried with no subcommands or arguments. Note that some responses, such as the SStatus command, change the number of lines generated based on device features and current configuration.

For example, ASet might respond with
Ch A POS ON Dly 00.000,000,000,000 Wid 00.000,002,000,000 50R, the
comprehensive active configuration of channel A.

For automated systems controlling the T660, it is recommended to use P500-style commands.

Verbose Mode

Since long strings of numbers are difficult to read, the T560 introduced *verbose mode*, which changes the way all response numbers are formatted. Verbose mode only affects T560-style sessions, and is a global, saved configuration property of the device.

In verbose mode, decimals are kept, and every three consecutive numeric digits a comma (,) character is printed. Responses affected include times, 32-bit integers, and frequencies. 16-bit integers are not affected.

For example, in verbose mode, the time 02.123456789123 becomes 02.123,456,789,123, and the 32-bit integer 0123456789 becomes 0,123,456,789.

T560-Style Commands List

Command Table

Below is a list of T560-style commands that can be executed while operating the T660 through a T560-style session. Not all commands are available to all T660 models; some commands or operations require the T660 to be purchased with the *Trains and Frames* feature.

Command	Models	Brief Description
Global Commands		
IDentify	All ²	Identify the device.
RSet	All	Reset the device.
FEod	All	Force an end-of-delay to occur.
Session Commands		
ZMode	All	Query the current command session type.
ZMode <mode>	All	Set the current command session type.
COmment	All	Ignore all characters until the next line delimiter.
WAit [<us>]	All	Wait a specified number of microseconds.
VErbose	All	Query the status of verbose reporting mode.
VErbose <mode>	All	Enable or disable verbose reporting mode.
VErbose ERrors	All	Query the session error verbosity level.
VErbose ERrors <level>	All	Set the session error verbosity level.
System Status Commands		
STatus	All ²	Procure a system status report.
ERrors	All ²	Query the error flags.
ERrors <n>	All	Clear the error flags.
USeC	All	Query the microsecond counter.
USeC <n>	All	Clear the microsecond counter.
IRq	All	Query the 40 Hz interrupt counter.
Channel Commands		
<A B C D>Pending	All	Query a channel's pending configuration.
<A B C D>Queued	T&F	Query a channel's queued configuration.
<A B C D>Set	All	Query a channel's active configuration.
<A B C D>Set ON	All	Enable a channel's output.
<A B C D>Set OFF	All	Disable a channel's output.
<A B C D>Set P0s	All	Set a channel's output to be active-high (non-inverted).
<A B C D>Set NEg	All	Set a channel's output to be active-low (inverted).
<A B C D>Set LOz	All	Disable source termination of an output channel connector.
<A B C D>Set TErminate	All	Enable source termination of an output channel connector.
Timing Commands		

Command	Mode Is	Brief Description
<A B C D>Delay	All	Query one or all channel's timing delay(s).
<A B C D Q>Delay <time>	All	Set one or all channel's timing delay(s).
<A B C D>Width	All	Query one or all channel's timing width(s).
<A B C D Q>Width <time>	All	Set one or all channel's timing width(s).
INstall	All	Apply pending timing settings immediately.
INstall <frame>	T&F	Apply a frame's timing settings immediately.
QQueue	T&F	Queue timing settings to be installed at next EOD.
QQueue <frame>	T&F	Queue a frame's timing settings to be installed at next EOD.
Autoinstall	All	Query the current autoinstall mode.
Autoinstall <mode>	All ³	Configure autoinstall functionality.
UNdo	All	Revert changes to the pending configuration.
Trigger Commands		
TLevel	All	Query the external trigger voltage threshold.
TLevel <volts>	All	Set the external trigger voltage threshold.
TRigger	All ²	Query the current trigger configuration.
TRigger P0s	All	Source triggers from external trigger rising edges.
TRigger NEg	All	Source triggers from external trigger falling edges.
TRigger INT	All	Source triggers from the internal 80 MHz clock.
TRigger SYn	All	Source triggers from the on-board frequency synthesizer.
TRigger REmote	All	Source triggers from remote trigger commands.
TRigger OFf	All	Disable (future) triggering.
TRigger HIz	All	Disable end termination of the trigger connector.
TRigger TErminate	All	Enable end termination of the trigger connector.
TDivider	All	Query the trigger pre-divider divisor value.
TDivider <divisor>	All	Set the trigger pre-divider divisor value.
TFreq	All	Query the trigger frequency counter.
TFreq INput	All	Set the trigger frequency counter to count all triggers received.
TFreq HIts	All	Set the trigger frequency counter to count hits only.
TPeriod	All	Query the trigger period, in nanoseconds.
FIre	All	Fire a remote trigger.
SHots	All	Query the shot counter.
SHots <n>	All	Clear the shot counter.
SYnthesize	All	Query the on-board trigger synthesizer frequency.
SYnthesize <frequency>	All	Set the frequency of the on-board trigger synthesizer.
Gate Commands		

Command	Mode Is	Brief Description
GAte	All	Query the current gate configuration.
GAte OFF	All	Disable gate functions.
GAte OUTput	All	Utilize the gate connector as an output.
GAte INput	All	Utilize the gate connector as an input.
GAte AUX	T&F	Use the gate connector to output the auxiliary output.
GAte POS	All	Utilize the gate connector as an active-high (normal) input/output.
GAte NEg	All	Utilize the gate connector as an active-low (inverted) input/output.
GAte TERminate	All	Enable gate connector end-termination while the gate connector is an input.
GAte HIz	All	Disable gate connector end-termination while the gate connector is an input.
GAte BURst	All	Clear the burst counter and disable bursts while the external gate input is low.
GAte REMote	All	Disable the gate connector and fire bursts with the GAte FIRE command.
GAte FIRE	All	Fire a single remote burst.
Auxiliary Output Commands		
AX	T&F	Query the configuration of the auxiliary output.
AX UPdate	T&F	Configure the auxiliary output to update mode.
AX TRigger	T&F	Configure the auxiliary output to trigger mode.
AX PWidth	T&F	Query the auxiliary output update mode pulse width.
AX PWidth <width>	T&F	Set the auxiliary output update mode pulse width.
Burst Commands		
BURst	All	Query the current burst configuration.
BURst ON	All	Enable burst mode.
BURst OFF	All	Disable burst mode.
BURst REset	All	Reset the burst counter.
BNum	All	Query the burst pulse count.
BNum <n_pulses>	All	Set the burst pulse count.
BMod	All	Query the burst period count.
BMod <m_triggers>	All	Set the burst period count.
Clock Commands		
CLock	All	Query clock settings and board temperature.
CLock HIz	All	Disable the clock connector.
CLock OUT	All	Output a 10 MHz clock via the clock connector.
CLock IN	All	Utilize the clock connector as an input and lock to an external 10 MHz clock.

Command	Mode Is	Brief Description
CLock SAve	All	Save the user-set clock trim value to nonvolatile memory.
CTrim	All	Query the VCXO trim value.
CTrim <trim>	All	Set the VCXO trim value.
Trains Commands		
TCount	T&F	Query the number of pulses per trigger.
TCount OFF	T&F	Disable pulse trains.
TCount <pulses>	T&F	Set the number of pulses per trigger.
TSpace	T&F	Query the pulse spacing.
TSpace <time>	T&F	Set the train pulse spacing.
Frames Commands		
FRame	T&F	Report the frames engine status.
FRame OFF	T&F	Cancel frame mode.
FRame <frame>	T&F	Store the current settings into the specified frame.
FRame GO	T&F	Start frames operation.
FRame LAsT	T&F	Report the last available frame index.
FDump	T&F	Display the current frame's contents.
FDump <frame>	T&F	Display a frame's contents.
FA	T&F	Query the start frame index.
FA <frame>	T&F	Set the start frame index.
FB	T&F	Query the end frame index.
FB <frame>	T&F	Set the end frame index.
FC	T&F	Query the frames loop count.
FC <count>	T&F	Set the frames loop count.
FN	T&F	Query the number of frames loaded into hardware.
FN <count>	T&F	Clear the loaded frame count.
FM	T&F	Query the number of frames remaining to be loaded.
FP	T&F	Query the current frame pointer.
FX	T&F	Report frames engine diagnostics.
RTest	T&F	Test the frames memory.
RZap	T&F	Clear the frames memory.
Memory Commands		
SAve	All	Save the current configuration to nonvolatile memory.
REcall	All	Load the system configuration from nonvolatile memory.
LOad DEfault	All	Load configuration defaults.
RUn DEMo	All	Load the demonstration configuration.
Network Commands		

Command	Models	Brief Description
SBAud	All	Query the serial port baud rate.
SBAud <baud>	All	Set the serial port baud rate.
NEtwork	All	Report the network status and active configuration.
NEtwork ADdress <ipv4_address>	All	Set the IPv4 address for the pending network configuration.
NEtwork ADdress	All	Query the IPv4 address set by the active network configuration.
NEtwork APply	All	Apply and save the pending network configuration.
NEtwork H0stname	All	Query the device hostname.
NEtwork MAC	All	Query the device MAC address.
NEtwork SUBnet <ipv4_subnet>	All	Set the IPv4 subnet for the pending network configuration.
NEtwork SUBnet	All	Query the IPv4 subnet set by the active network configuration.
Help Commands		
HElp	All ²	Display the help commands help page.
HElp CHannels	All	Display the channel and timing commands help page.
HElp TRigger	All	Display the trigger commands help page.
HElp GAtE	All	Display the gate commands help page.
HElp BURst	All	Display the burst commands help page.
HElp CLock	All	Display the clock commands help page.
HElp MISC	All	Display the misc. commands help page.
HElp DEMo	All	Display the demo mode help page.
HElp PULsetrain	T&F	Display the trains commands help page.
HElp FRames	T&F	Display the frames commands help page.

2(1, 2, 3, 4, 5) The behavior of this command changes depending on the model.

3 Some features of this command are only available to *Trains and Frames* models.

Global Commands

Identify

Description: Identify the device.

Reports The model and firmware version of the device. To preserve T560 compatibility, this command does not report the device serial number; the SStatus command reports this.

Response: Responds with T660 DDG Firmware <firmware_version>, where <firmware_version> is the firmware version of the device. For example, the first-released version number is read as 28E660-1-1.0.

RSet

Description: Reset the device.

Immediately, upon execution, performs a soft reboot of the device. This has the side-effect of resetting all network connections. All unsaved configuration will be lost.

If for some reason this command fails to execute properly, its default behavior is to produce an error (specifically, a ?11: Software error).

Response: This command should never produce a response because the device should reboot before it does. However, at the initiation of a T560-style session, the T660 prints a newline delimiter, followed by "Highland Technology T660 DDG", followed by another newline delimiter; this can be used to determine that the device has rebooted and is ready to accept commands.

FEod

Description: Force an end-of-delay to occur.

Forces an end-of-delay, terminating any possibly-running shot. This applies any queued configuration settings, even if a shot was not in progress.

Response: OK or ?? (to indicate an error).

Notes:

- When run while the frames engine is active, this command advances frames engine operation as though a shot completed normally, even if a shot was not in progress.
- As of firmware v1.7, for compatibility, this command is equivalent to issuing a FRame OFF command *only when run via the isochronous interface*.

Session Commands

ZMode

Description: Query the current command session type.

Response: Either P500 or T560 to indicate which command mode the current session is currently operating as.

Notes:

- In T560-style sessions, a trailing ? is unnecessary. However, it is recommended to always use a trailing ? character because this syntax is compatible with both P500- and T560-style sessions.

ZMode <mode>

Description: Set the current command session type.

Gets or sets the current session's command parsing mode. Upon successfully changing the command parsing mode, terminates execution of the current command line immediately. This means that this command, as a command, should not have any commands following it on a single command line!

Parameters:

- <mode> - the mode to set the current command session to, one of {P500, T560}. Note that only the first two characters of each value is required.

Response: OK, or an ?07: Illegal operation error if the session type was unable to be changed.

Notes:

- Not all command sessions can switch between parsing modes. In these cases, this command will produce a ?07: Illegal operation error.
- If the command session mode was not changed, this command does not terminate the current line after execution and yields an OK response.

COmment

Description: Ignore all characters until the next line delimiter.

Stops parsing commands for the remainder of the current command line.

Response: There is no response provided by this command. This is not to be confused with a blank response.

Notes:

- This command will immediately terminate execution of the remainder of the line.
- Command lines ending with a COmment command do not apply autoinstalls.
- Pending autoinstalls are not cleared by this command, and are forwarded to the next command line.
- Due to the above notes, this command is not equivalent to an 'empty' command consisting of no characters.

WAit [<us>]

Description: Wait a specified number of microseconds.

This command is not intended to be timing accurate, and in many cases is only accurate to within approximately 100 microseconds. The accuracy of this command will depend on other concurrent sessions, processor loading, and various other factors.

Parameters:

- <us> - The number of microseconds to wait, as an unsigned 32-bit integer.

Response: OK, after the wait has completed.

Notes:

- While a wait command is executing, other sessions are allowed to execute commands.

VErbose

Description: Query the status of verbose reporting mode.

Response: The T560-style numerics verbosity mode, as a 16-bit unsigned integer. 00000 for disabled, 00001 for enabled.

Notes:

- Verbose mode is configured globally for the device, not per session, and only applies in T560-style sessions.
- Verbose mode is a saved (non-volatile) configuration parameter, loaded at device power-up, and saved by the SAve command.

VErbose <mode>

Description: Enable or disable verbose reporting mode.

When verbose mode is enabled, commas are inserted into all reported 32-bit unsigned integers, timing configuration values, and frequencies at groupings of 3 digits, centered at the decimal point (express or implied). Verbose mode does not affect the serialization of 16-bit unsigned integers (those are still formatted as 00000) despite applying to 32-bit unsigned integers (0,000,000,000 when enabled, 0000000000 when disabled). This configuration option affects the responses of various commands.

Parameters:

- <mode> - The numerics verbosity mode, as a 16-bit unsigned integer. 0 for disabled, 1 for enabled. All non-zero values will be interpreted as enabled (1).

Response: OK or ?? (to indicate an error).

Notes:

- Verbose mode is configured globally for the device, not per session, and only applies in T560-style sessions.
- Verbose mode is a saved (non-volatile) configuration parameter, loaded at device power-up, and saved by the SAve command.

VERbose ERRors**Description:** VERbose ERRors**Response:** Responds with the current session's error verbosity setting, as a 16-bit unsigned integer (0 through 3, inclusive). See the VERbose ERRors <level> command for definitions. Values greater than 3 may be returned if they were set with the VERbose ERRors <level> and represent the highest available verbosity level.**VERbose ERRors <level>****Description:** VERbose ERRors <level>

Sets the error verbosity level of the current session, leaving other sessions unchanged. This command takes effect immediately and affects all future commands, even those in the same command line.

When an error occurs as a result of any command in the current session, diagnostic information about the error is sent as a response instead of the command's normal response. This information may include:

1. At least one question mark character (?)
2. The error code, as a two digit decimal integer
3. The error name
4. A brief description of the error

Error verbosity is represented as an integer, lower values representing less information. A table of verbosity levels is provided below:

Level	Information	Example
0	The text ??	??
1	Error code	?32
2	Error code and name	?32: Unexpected argument
3	Error code, name, and brief description	?32: Unexpected argument: Too many arguments provided for the specified command.

Parameters:

- <level> - The verbosity level to set, as an integer. Only values 0 through 3 (inclusive) are valid. Values higher than 3 are interpreted as the highest available level.

Response: OK or ?? (to indicate an error).

Notes:

- Session verbosity is a per-session configuration; multiple sessions may have different values, and configuration does not persist across reboots.
- When session mode is changed via the ZMode <mode> command, error verbosity is reset to its default value for the new session type (0 for T560-style, and 3 for P500-style), so this command may need to be re-run after session type changes.
- P500-style sessions default to an error verbosity of 3, while T560-style sessions default to a verbosity of 0.

System Status Commands

Status

Description: Procure a system status report.

Reports device-unique identifying information, followed by the configuration state of the device. To preserve T560 compatibility, this command does not report the specific firmware version; the IDentify command reports this.

Response: Responds with a large, multi-line description of the device state, including the configuration of all four output channels. The displayed configuration is the active configuration, not the pending or queued ones.

All data apart from the header of the response can be generated using various other status commands.

Notes:

- changes based on the availability of the frames and trains feature on the device.
- This command produces a response that is multiple lines long (the response includes line-endings).

ERrors

Description: Query the error flags.

Response: Responds with the global errors flags as a bitmask of a 16-bit unsigned integer, if there are any errors, followed by a friendly abbreviated name for each error that exists. This means that the length of the response from this command can vary substantially depending on the number of error flags asserted.

The format of the response is Errs None if there are no errors present, or Errs <mask> <list> if any error flags are asserted. Errors are printed in the order defined by their bitfield representation, with bit 0 appearing first, and bit 7 appearing last. Errors, their bit positions (0 to 15), and their abbreviated names as reported by this command are:

Bit	Name	Description
0	XTRIM	Oscillator trim failure.
1	RECALL	Failed to recall saved configuration. May occur at system power-on or by execution of the RECALL command.
2	CALIB	Invalid or lost calibration. May occur at system power-on.
3	LOGIC	Internal logic error. Asserted if the software ever encounters an internal error resulting in a potentially unsafe or undefined system state.
4	VCXO	VCXO lock error (self-clearing). Indicates a failure to lock to an external 10 MHz clock.
5	TUNE	Oscillator startup tune error. Indicates that the shot oscillator failed to calibrate at device power-on.
6	DPLL	Oscillator runtime tune error. Indicates that the shot oscillator was unable to compensate for real-time frequency variation. This is typically caused by an out-of-spec 10 MHz reference frequency or extreme ambient temperatures.
7	POWER	Power supply monitoring error. Indicates that the background built-in-self-test (BIST) function measured one or more on-board power supplies as beyond allowed voltage tolerance. This can indicate a hardware failure, or be caused by extreme ambient temperatures.
8	OLOAD	Channel output overload. Asserted while the front-panel channel output connectors are connected to a load that is drawing too much current, forcing them into a protection state.
9	TRATE	Trigger rate warning. Asserted for a short time while any triggers are received that would otherwise cause a shot to occur, but a shot was already in progress.
10	TEMP0	Active temperature compensation of the internal oscillator failed due to the temperature being measured as outside of the correctable range.
11	TEMP1	Active temperature compensation of insertion delay failed due to the temperature being measured as outside of the correctable range.

Errors <n>

Description: Clear the error flags.

Clears all error flags reported by the ERRors command. Persistent errors, such as the VCXO lock flag, while cleared temporarily, will be re-asserted after an undefined amount of time (less than one second) if they persist.

Parameters:

- <n> - May be any 32-bit integral number. This number is ignored.

Response: OK or ?? (to indicate an error).

Usec

Description: Query the microsecond counter.

Resets the microsecond counter to 0 upon execution.

Response: Responds with the number of microseconds that have elapsed since the microsecond counter was last cleared or the device was last powered on or rebooted, formatted as an unsigned 32-bit integer.

Usec <n>

Description: Clear the microsecond counter.

Parameters:

- <n> - May be any 32-bit integral number. This number is ignored.

Response: OK or ?? (to indicate an error).

IRq

Description: Query the 40 Hz interrupt counter.

Response: Responds with the number of 40 Hz intervals that have elapsed since the device was last powered on or rebooted, formatted as an unsigned 16-bit integer.

Notes:

- This command is deprecated, and is provided solely for T560 compatibility.

Channel Commands

<A|B|C|D>Pending

Description: Query a channel's pending configuration.

Reports the full pending (not active) configuration of the channel specified by the first character of this command (A, B, C, or D). This includes, specifically: channel name, polarity, enabled state, timing configuration (delay and width), and output termination.

Response: Responds with text representing the pending configuration for the channel requested, in the same format as the <A|B|C|D>Set command.

Notes:

- The response of this command is affected by verbose mode.

<A|B|C|D>Queued

Description: Query a channel's queued configuration.

Reports the full queued (not active) configuration of the channel specified by the first character of this command (A, B, C, or D). This includes, specifically: channel name, polarity, enabled state, timing configuration (delay and width), and output termination.

Response: Responds with text representing the queued configuration for the channel requested, in the same format as the <A|B|C|D>Set command.

Notes:

- This command is operable while the Frames engine is running. However, since the frames engine operates asynchronously to commands, channel and timing configuration may change while this command is being executed, causing undefined behavior. Therefore, it is not recommended to use this command while the Frames engine is operational, although using it while frames are guaranteed not to change is supported.
- The response of this command is affected by verbose mode.

<A|B|C|D>Set

Description: Query a channel's active configuration.

Reports the full active (not pending) configuration for the channel specified by the first character of this command (A, B, C, or D). This includes, specifically: channel name, polarity, enabled state, timing configuration (delay and width), and output termination.

Response: Responds with text representing the active configuration for the channel requested. This text takes the form: Ch <channel> <polarity> <enabled> Dly <delay> Wid <width> <termination>, where:

- <channel> may be one of {A, B, C, D}
- <polarity> may be one of {POS, NEG}
- <enabled> may be one of {ON, OFF}
- <delay> is the rising edge delay, in seconds, in the same format as the <A|B|C|D>DeLay command
- <width> is the pulse width, in seconds, in the same format as the <A|B|C|D>Width command
- <termination> may be one of {50R, LOZ}

Notes:

- This command is operable while the Frames engine is running. However, since the frames engine operates asynchronously to commands, channel and timing configuration may change while this command is being executed, causing undefined behavior. Therefore, it is not recommended to use this command while the Frames engine is operational, although using it while frames are guaranteed not to change is supported.
- The response of this command is affected by verbose mode.

<A|B|C|D>Set ON

Description: Enable a channel's output.

Modifies the pending configuration for the channel specified by the first character of this command (A, B, C, or D) and queues an autoinstall.

Response: OK or ?? (to indicate an error).

Notes:

- This command modifies the pending output configuration, and requires an autoinstall, install, or queue to take effect.
- This command will set the autoinstall flag when executed.

<A|B|C|D>Set OFF

Description: Disable a channel's output.

Modifies the pending configuration for the channel specified by the first character of this command (A, B, C, or D) and queues an autoinstall.

Response: OK or ?? (to indicate an error).

Notes:

- This command modifies the pending output configuration, and requires an autoinstall, install, or queue to take effect.
- This command will set the autoinstall flag when executed.

<A|B|C|D>Set P0s

Description: Set a channel's output to be active-high (non-inverted).

Modifies the pending configuration for the channel specified by the first character of this command (A, B, C, or D) and queues an autoinstall.

Response: OK or ?? (to indicate an error).

Notes:

- This command modifies the pending output configuration, and requires an autoinstall, install, or queue to take effect.
- This command will set the autoinstall flag when executed.

<A|B|C|D>Set NEg

Description: Set a channel's output to be active-low (inverted).

Modifies the pending configuration for the channel specified by the first character of this command (A, B, C, or D) and queues an autoinstall.

Response: OK or ?? (to indicate an error).

Notes:

- This command modifies the pending output configuration, and requires an autoinstall, install, or queue to take effect.
- This command will set the autoinstall flag when executed.

<A|B|C|D>Set L0z

Description: Disable source termination of an output channel connector.

Modifies the pending configuration for the channel specified by the first character of this command (A, B, C, or D) and queues an autoinstall.

Response: OK or ?? (to indicate an error).

Notes:

- This command modifies the pending output configuration, and requires an autoinstall, install, or queue to take effect.
- This command will set the autoinstall flag when executed.

<A|B|C|D>Set TErminate

Description: Enable source termination of an output channel connector.

Modifies the pending configuration for the channel specified by the first character of this command (A, B, C, or D) and queues an autoinstall.

Response: OK or ?? (to indicate an error).

Notes:

- This command modifies the pending output configuration, and requires an autoinstall, install, or queue to take effect.
- This command will set the autoinstall flag when executed.

Timing Commands

<A|B|C|D>Delay

Description: Query one or all channel's timing delay(s).

Reports the pending configuration rising edge delay of the channel specified by the first character of this command (A, B, C, or D).

Response: The rising edge delay of the specified channel, in seconds. This result is formatted 00.000,000,000,000 when verbose mode is 1, or 00.000000000000 when verbose mode is 0.

Notes:

- The response of this command is affected by verbose mode.

<A|B|C|D|Q>Delay <time>

Description: Set one or all channel's timing delay(s).

Overwrites the pending configuration rising edge delay of the channel specified by the first character of this command (A, B, C, or D).

Parameters:

- <time> - The desired rising edge delay. Available suffixes are Ps (picoseconds), Ns (nanoseconds), Ms (milliseconds), and S (seconds). Default units are nanoseconds. Scientific notation is accepted. Negative numbers are not allowed.

Response: OK or ?? (to indicate an error).

Notes:

- This command modifies the pending output configuration, and requires an autoinstall, install, or queue to take effect.
- This command will set the autoinstall flag when executed.

<A|B|C|D>Width

Description: Query one or all channel's timing width(s).

Reports the pending configuration pulse width of the channel specified by the first character of this command (A, B, C, or D).

Response: Responds with the pulse width of the specified channel, in seconds. This result is formatted 00.000,000,000,000 when verbose mode is 1, or 00.000000000000 when verbose mode is 0.

Notes:

- The response of this command is affected by verbose mode.

<A|B|C|D|Q>Width <time>

Description: Set one or all channel's timing width(s).

Overwrites the pending configuration pulse width of the channel specified by the first character of this command (A, B, C, or D).

Parameters:

- <time> - The desired pulse width. Available suffixes are Ps (picoseconds), Ns (nanoseconds), Ms (milliseconds), and S (seconds). Default units are nanoseconds. Scientific notation is accepted. Negative numbers are not allowed.

Response: OK or ?? (to indicate an error).

Notes:

- This command modifies the pending output configuration, and requires an autoinstall, install, or queue to take effect.
- This command will set the autoinstall flag when executed.

INStall

Description: Apply pending timing settings immediately.

Queues the pending timing configuration (consisting of channel configuration, timing settings, and pulse trains configuration) to be applied to the active configuration, then forces an immediate end-of-delay, terminating any possible running shot and applying the queued configuration. Issuing this command is functionally equivalent to issuing a QUeue command followed by a FEod command.

Response: OK or ?? (to indicate an error).

Notes:

- This command modifies the queued configuration.
- Running this command terminates any possibly-running shot.
- Running this command clears a pending autoinstall, regardless of autoinstall mode.

INStall <frame>

Description: Apply a frame's timing settings immediately.

Overwrites the pending timing configuration with the contents of the specified frame, (consisting of channel configuration, timing settings, and pulse trains configuration), then issues an install (functionally equivalent to issuing a INStall command). Issuing this command is functionally equivalent to issuing a QUeue <frame> command followed by a FEod command.

Parameters:

- <frame> - The index of the frame to load configuration from.

Response: OK or ?? (to indicate an error).

Notes:

- This command modifies the pending and queued configurations.
- This command terminates any possibly-running shot.
- Running this command clears a pending autoinstall, regardless of autoinstall mode.

QUeue

Description: Queue timing settings to be installed at next EOD.

Queues the pending timing configuration (consisting of channel configuration, timing settings, and pulse trains configuration) to be applied to the active configuration at the next end-of-delay.

Response: OK or ?? (to indicate an error).

Notes:

- This command modifies the queued configuration, and requires an EOD (manual or at the end of a shot) to take effect.
- Running this command clears a pending autoinstall, regardless of autoinstall mode.

QUeue <frame>

Description: Queue a frame's timing settings to be installed at next EOD.

Overwrites the pending timing configuration with the contents of the specified frame, (consisting of channel configuration, timing settings, and pulse trains configuration), then issues a queue (functionally equivalent to issuing a QUeue command).

Parameters:

- <frame> - The index of the frame to load configuration from.

Response: OK or ?? (to indicate an error).

Notes:

- This command modifies the queued configuration, and requires an EOD (manual or at the end of a shot) to take effect.
- Running this command clears a pending autoinstall, regardless of autoinstall mode.

AUTOinstall

Description: Query the current autoinstall mode.

Response: Responds with the currently-configured autoinstall mode, as a 16-bit unsigned integer. This may be one of three values, where:

- 00000 indicates that autoinstall is disabled.
- 00001 indicates that an INStall operation will be executed at the end of a command line, whenever an autoinstall has been queued for that line.
- 00002 indicates that a QUeue operation will be executed at the end of a command line, whenever an autoinstall has been queued for that line.

AUTOinstall <mode>

Description: Configure autoinstall functionality.

Configure the automatic application of the pending configuration at the end of command lines. Autoinstall may be configured to one of three modes:

- 0 - Autoinstall is disabled.
- 1 - An INStall operation will be executed at the end of a command line, whenever an autoinstall has been queued for that line.
- 2 - A QUeue operation will be executed at the end of a command line, whenever an autoinstall has been queued for that line.

Autoinstall can only be configured to queue mode (mode 2) if the trains and frames feature is accessible.

Parameters:

- <mode> - The mode to configure autoinstall to, as a 16-bit unsigned integer. May be 0 (off), 1 (install), or 2 (queue). Other values will be interpreted as queue mode.

Response: OK or ?? (to indicate an error).

Notes:

- Because unknown integer values are interpreted as mode 2, this command may produce an error if trains and frames is not accessible.

UNdo

Description: Revert changes to the pending configuration.

Loads the queued configuration into the pending configuration.

Response: OK or ?? (to indicate an error).

Notes:

- This command modifies the pending configuration, and requires a queue or install to take effect.
- This command does not affect autoinstall functionality, despite modifying the pending configuration.

Trigger Commands

TLevel

Description: Query the external trigger voltage threshold.

Response: Responds with the currently-configured external trigger logic level threshold, in Volts, in the format 0.000.

TLevel <volts>

Description: Set the external trigger voltage threshold.

Configures the external trigger connector's logic-level threshold. Voltages above this threshold are considered high, and those below low. This does not account for hysteresis.

Parameters:

- <volts> - The threshold voltage to set, in Volts, as a floating point number. No suffixes are supported. Values lower than 0.25 Volts and higher than 3.3009 Volts are not allowed and will produce an error.

Response: OK or ?? (to indicate an error).

TRigger

Description: Query the current trigger configuration.

Response: Trig <source> <termination> Level <voltage> Div <divisor> SYN <frequency>, where:

- <source> may be one of {OFF, INT, SYN, REM, POS, NEG} representing the configured trigger source. The values POS and NEG indicate that the trigger source is set to external, and indicate the configured external trigger polarity.
- <termination> may be either 50R or HIZ, indicating whether the trigger connector is end terminated.
- <voltage> is the external trigger level, in Volts, equal to the response generated by the TLevel command.
- <divisor> is the trigger predivider divisor value, equal to the response generated by the TDivider command.
- <frequency> is the configured trigger synthesizer frequency, in Hz, equal to the response generated by the SYNthesize command.

If the frames and trains feature is available, an additional line will be printed, preceded and succeeded by a newline delimiter. This line is of the format

<mode> Freq Hz <t_frequency> Period ns <t_period>, where:

- <mode> indicates whether the trigger frequency counter is measuring hits or triggers, either Hit if set to count hits, or Inp if set to count triggers.
- <t_frequency> is the frequency of the frequency counter in Hertz, equal to the response generated by the TFreq command.
- <t_period> is the period of the frequency counter in nanoseconds, equal to the response generated by the TPeriod command.

Notes:

- changes based on the availability of the frames and trains feature on the device.
- This command produces a response that is multiple lines long (the response includes line-endings) when the frames and trains feature is available.
- The response of this command is affected by verbose mode when the frames and trains feature is available.

TRigger P0s

Description: Source triggers from external trigger rising edges.

Changes the trigger source configuration to source triggers only from the external trigger connector. Even if the current trigger source is already identical to the one requested, execution of this command will temporarily disable triggers, so triggers may be missed if repeatedly issued.

Response: OK or ?? (to indicate an error).

TRigger NEg

Description: Source triggers from external trigger falling edges.

Changes the trigger source configuration to source triggers only from the external trigger connector. Even if the current trigger source is already identical to the one requested, execution of this command will temporarily disable triggers, so triggers may be missed if repeatedly issued.

Response: OK or ?? (to indicate an error).

TRigger INT

Description: Source triggers from the internal 80 MHz clock.

Changes the trigger source configuration to source triggers only from the internal 80 MHz oscillator. Even if the current trigger source is already identical to the one requested, execution of this command will temporarily disable triggers, so triggers may be missed if repeatedly issued.

Response: OK or ?? (to indicate an error).

TRigger SYn

Description: Source triggers from the on-board frequency synthesizer.

Changes the trigger source configuration to source triggers only from the on-board trigger frequency synthesizer. Even if the current trigger source is already identical to the one requested, execution of this command will temporarily disable triggers, so triggers may be missed if repeatedly issued.

Response: OK or ?? (to indicate an error).

TRigger REMote

Description: Source triggers from remote trigger commands.

Changes the trigger source configuration to source triggers only from execution of the FIRE command. Even if the current trigger source is already identical to the one requested, execution of this command will temporarily disable triggers, so triggers may be missed if repeatedly issued.

Response: OK or ?? (to indicate an error).

TRigger OFF

Description: Disable (future) triggering.

Disables triggers by sourcing triggers from a logic-low source. This does not terminate any possibly-running shot, so applying this configuration change does not necessarily indicate that future outputs will be disabled.

Response: OK or ?? (to indicate an error).

TRigger HIz

Description: Disable end termination of the trigger connector.

Disengage the 50 Ohm (to ground) end-termination of the external trigger connector.

Response: OK or ?? (to indicate an error).

TRigger TErminate

Description: Enable end termination of the trigger connector.

Engage the 50 Ohm (to ground) end-termination of the external trigger connector.

Response: OK or ?? (to indicate an error).

TDivider

Description: Query the trigger pre-divider divisor value.

Reports the number of triggers that should be received before the next trigger is considered valid. A value of 0 is functionally equivalent to a value of 1, and indicates that the predivider is disabled.

Response: Responds with the configured predivider divisor, as a 32-bit unsigned integer.

Notes:

- The response of this command is affected by verbose mode.

TDivider <divisor>

Description: Set the trigger pre-divider divisor value.

Sets the number of triggers that should be received before the next trigger is considered valid. A value of 0 is functionally equivalent to a value of 1, and indicates that the predivider is disabled. This configuration applies to all trigger sources, even remote triggers.

Parameters:

- <divisor> - The trigger divisor value to set, as a 32-bit unsigned integer. Hexadecimal notations are allowed, and the full range of the integer is valid. 0 implies a divisor value of 1.

Response: OK or ?? (to indicate an error).

TFreq

Description: Query the trigger frequency counter.

Reports the trigger frequency counter's frequency measurement, running averaged over the course of 1 second

Response: Responds with the frequency counter's frequency measurement, in Hertz, as a 32-bit unsigned integer.

Notes:

- The response of this command is affected by verbose mode.
- Due to a 1 second running average measurement delay, the response of this command may not immediately represent the state of the hardware.
- Periods of inactivity are not counted towards the running average. As such, immediate changes in the incoming trigger signal may produce immediate results earlier than 1 second later. This also means that averaging trigger rates over periods of intermittent trigger inactivity (such as those longer than 8 milliseconds) may yield incorrectly averaged measurements.

TFreq Input

Description: Set the trigger frequency counter to count all triggers received.

Configures the trigger frequency counter to measure incoming triggers, as opposed to hits.

Response: OK or ?? (to indicate an error).

Notes:

- Because the T660 always measures both trigger frequency counter modes, although this command affects the responses of the TFreq and TPeriod commands, their values will reflect the appropriate measurement immediately and do not require additional settling time.

TFreq Hits

Description: Set the trigger frequency counter to count hits only.

Configures the trigger frequency counter to measure hits, as opposed to incoming triggers.

Response: OK or ?? (to indicate an error).

Notes:

- Because the T660 always measures both trigger frequency counter modes, although this command affects the responses of the TFreq and TPeriod commands, their values will reflect the appropriate measurement immediately and do not require additional settling time.

TPeriod

Description: Query the trigger period, in nanoseconds.

Reports the trigger frequency counter's period measurement, running averaged over the course of 1 second

Response: Responds with the trigger frequency counter's period measurement, in nanoseconds, as a 32-bit unsigned integer.

Notes:

- The response of this command is affected by verbose mode.
- Due to a 1 second running average measurement delay, the response of this command may not immediately represent the state of the hardware.
- Periods of inactivity are not counted towards the running average. As such, immediate changes in the incoming trigger signal may produce immediate results earlier than 1 second later. This also means that averaging trigger rates over periods of intermittent trigger inactivity (such as those longer than 8 milliseconds) may yield incorrectly averaged measurements.

Fire

Description: Fire a remote trigger.

Executes a remote trigger only if the trigger source is set to remote. If it is not, this command produces an error (specifically, a ?07: Illegal operation error).

Response: OK or ?? (to indicate an error).

Notes:

- This command can only be executed with certain global configurations, and has normal behavior that includes producing a ?07: Illegal operation error if this requirement is not met.
- Remote triggers are affected by predivider and burst configuration just like other trigger sources.

SHots

Description: Query the shot counter.

Response: Responds with the number of shots that have elapsed since SHots <n> was last called or the device was last powered on or rebooted, formatted as an unsigned 32-bit integer.

SHots <n>

Description: Clear the shot counter.

Parameters:

- <n> - May be any integral number. This number is ignored.

Response: OK or ?? (to indicate an error).

SYnthesize

Description: Query the on-board trigger synthesizer frequency.

Reports the configured trigger synthesizer frequency. This value is not measured, but instead reported from configuration.

Response: The configured trigger synthesizer frequency, in Hertz, formatted as 00000000.00 when verbose mode is disabled, or 00,000,000.00 when verbose mode is enabled.

Notes:

- The response of this command is affected by verbose mode.

SYnthesize <frequency>

Description: Set the frequency of the on-board trigger synthesizer.

The maximum synthesizable frequency is 16 MHz; values higher than this will produce an error.

Parameters:

- <frequency> - The frequency to set the trigger synthesizer to, in Hertz by default, as a floating-point number. Available suffixes are Mhz, Khz, and Hz. Values higher than 16 MHz or lower than 0 Hz (negative values) are not allowed.

Response: OK or ?? (to indicate an error).

Notes:

- Although unlikely in practice, frequency transitions are not guaranteed to be glitchless; it is possible for the trigger rate to intermediately be higher or lower than both the previous and new frequency values temporarily.
- For extremely low frequencies, it is recommended to utilize the trigger predivider in combination with the internal trigger source to reduce potentially significant jitter (see: TDivider <divisor>).

Gate Commands

GATe

Description: Query the current gate configuration.

Response: Responds with the current gate configuration, formatted in the form Gate <mode> <polarity> <termination>. If termination is disabled for the gate connector, the reply is of the format Gate <mode> <polarity> <termination> Shots <shots> instead. For both of these formats:

- <mode> is one of OFF (off), OUT (output), INP (input), BUR (gate burst), or REM (remote), indicating the current gate operating mode.
- <polarity> is either POS or NEG, reflecting the I/O polarity set by the GATe POS and GATe NEg commands.
- <termination> is either HIZ or 50R, reflecting the termination state set by the GATe TErminate and GATe HIZ commands.
- <shots>, if present, reports the number of hits that have elapsed since system power-on as an unsigned 32-bit integer, equivalent to the response generated by the SHots command.

Notes:

- Unintuitively, the Shots <shots> portion of the response generated by this command is dependent on the termination state of the gate connector. This behavior is to preserve T560 compatibility.
- The response of this command is affected by verbose mode when the gate connector termination is disabled.

GATe OFF

Description: Disable gate functions.

Configures the gate connector operating mode to be gate off mode. This disables all gate functionality, but engages the configured gate input termination (if configured; see: the GATe TErminate and GATe HIZ commands).

Response: OK or ?? (to indicate an error).

Notes:

- Execution of this command clears *both* the predivider and burst counters.
- This command terminates any possibly-running shot by issuing an end-of-delay (see: FEod).

GATe Output

Description: Utilize the gate connector as an output.

Configures the gate connector operating mode to be gate output mode. In this mode, the the gate connector is used as an output. A logic-high indicates that the next incoming trigger will start a shot, while a logic-low indicates the opposite.

In modes the gate connector is utilized as an output, end termination is disabled. A 50-Ohm source termination is always present in this mode.

Response: OK or ?? (to indicate an error).

Notes:

- Execution of this command clears *both* the predivider and burst counters.
- This command terminates any possibly-running shot by issuing an end-of-delay (see: FEod).

GAte INput

Description: Utilize the gate connector as an input.

Configures the gate connector operating mode to be gate input mode. In this mode, the gate connector determines whether triggers are allowed to start shots. This is only allowed to happen if the gate input is logic-high.

Response: OK or ?? (to indicate an error).

Notes:

- Execution of this command clears *both* the predivider and burst counters.
- This command terminates any possibly-running shot by issuing an end-of-delay (see: FEod).

GAte AUx

Description: Use the gate connector to output the auxiliary output.

Configures the gate connector to be used as an output for the auxiliary output (see [Auxiliary Output Commands](#)). Normal gate functions behave as though the gate is off. In this mode, the gate polarity configuration is still used to determine the output signal level of the gate connector.

In modes the gate connector is utilized as an output, end termination is disabled. A 50-Ohm source termination is always present in this mode.

Response: OK or ?? (to indicate an error).

Notes:

- Execution of this command clears *both* the predivider and burst counters.
- This command terminates any possibly-running shot by issuing an end-of-delay (see: FEod).

GAte POs

Description: Utilize the gate connector as an active-high (normal) input/output.

Configures the gate connector to operate using normal (active-high) logic levels in all gate modes (input and output).

Response: OK or ?? (to indicate an error).

Notes:

- Executing this command may cause an unintended immediate gate burst if the gate mode is set to gate burst mode, since the logic level will be perceived as changing.

GAte NEg

Description: Utilize the gate connector as an active-low (inverted) input/output.

Configures the gate connector to operate using inverted (active-low) logic levels in all gate modes (input and output).

Response: OK or ?? (to indicate an error).

Notes:

- Executing this command may cause an unintended immediate gate burst if the gate mode is set to gate burst mode, since the logic level will be perceived as changing.

GAte TErminate

Description: Enable gate connector end-termination while the gate connector is an input.

Enables end-termination of the gate connector while not an output. Use the GATe HIz command to disable termination.

Response: OK or ?? (to indicate an error).

Notes:

- Termination configuration applies only when the gate mode is not set to output. This means that this configuration applies even while in gate remote mode and while gate functionality is configured as off.
- This command has no effect on hardware when run while the gate connector is utilized as an output (in output or auxiliary modes). However, the configured setting is applied immediately and will take effect once the gate connector is no longer utilized as an output.

GATe HIz

Description: Disable gate connector end-termination while the gate connector is an input.

Disables end-termination of the gate connector while not an output. Use the GATe TERminate command to enable termination.

Response: OK or ?? (to indicate an error).

Notes:

- Termination configuration applies only when the gate mode is not set to output. This means that this configuration applies even while in gate remote mode and while gate functionality is configured as off.
- This command has no effect on hardware when run while the gate connector is utilized as an output (in output or auxiliary modes). However, the configured setting is applied immediately and will take effect once the gate connector is no longer utilized as an output.

GATe BURst

Description: Clear the burst counter and disable bursts while the external gate input is low.

Configures the gate connector operating mode to be gate burst mode. In this mode, the rising edge of the gate input (logical low to logical high transition, as determined by the gate polarity) initiates the start of a burst sequence starting with the next trigger.

Response: OK or ?? (to indicate an error).

Notes:

- Execution of this command clears *both* the predivider and burst counters.
- This command terminates any possibly-running shot by issuing an end-of-delay (see: FEod).
- Operating with the gate operation set to remote or gate burst mode with burst disabled (invalid N and M values satisfy this) behaves as though burst is enabled, with N = 1 and M = 1.

GATe REMote

Description: Disable the gate connector and fire bursts with the GATe FIRE command.

Configures the gate connector operating mode to be gate remote mode. In this mode, the gate functionality operates similarly to gate burst mode (see: GATe BURst). Execution of the GATe FIRE command initiates the start of a burst sequence starting with the next trigger.

This mode utilizes the gate connector as an output. In modes the gate connector is utilized as an output, end termination is disabled. A 50-Ohm source termination is always present in this mode.

Response: OK or ?? (to indicate an error).

Notes:

- Execution of this command clears *both* the predivider and burst counters.
- This command terminates any possibly-running shot by issuing an end-of-delay (see: FEod).
- Operating with the gate operation set to remote or gate burst mode with burst disabled (invalid N and M values satisfy this) behaves as though burst is enabled, with N = 1 and M = 1.

Gate Fire

Description: Fire a single remote burst.

Simulates the gate connector pulsing high to fire a remote burst while the device is in gate remote mode.

Response: OK or ?? (to indicate an error).

Notes:

- If this command is run while the device is not in gate remote mode (see: GATe REmote), it will still produce the response OK, but will not perform any function.

Auxiliary Output Commands

AX

Description: Query the configuration of the auxiliary output.

Response: Responds with Aux Mode <mode> Width <width>, where:

- <mode> is either UPDATE or TRIGGER, referring to auxiliary output update and trigger modes, respectively.
- <width> is the width of pulses generated in update mode, in seconds. This result is formatted 00.000,000,000,000 when verbose mode is 1, or 00.000000000000 when verbose mode is 0. This value is not used in trigger mode.

AX UPdate

Description: Configure the auxiliary output to update mode.

Response: OK or ?? (to indicate an error).

Notes:

- Because the T660 does not have a dedicated auxiliary output connector, this command has no visible effect on hardware unless the gate connector has been configured to output the auxiliary output using the GATe AUx command.

AX TRigger

Description: Configure the auxiliary output to trigger mode.

Response: OK or ?? (to indicate an error).

Notes:

- Because the T660 does not have a dedicated auxiliary output connector, this command has no visible effect on hardware unless the gate connector has been configured to output the auxiliary output using the GAtE AUx command.

AX PWidth

Description: Query the auxiliary output update mode pulse width.

Response: Responds with the currently-configured width of pulses generated in update mode, in seconds. This result is formatted 00.000,000,000,000 when verbose mode is 1, or 00.000000000000 when verbose mode is 0.

AX PWidth <width>

Description: Set the auxiliary output update mode pulse width.

Sets the width of pulses generated by the auxiliary output when in update mode. The auxiliary output, when in trigger mode, is not affected by this setting.

Parameters:

- <width> - The update mode pulse width. Values 0 up to 3ms (inclusive) are allowed. This value is rounded down to the nearest multiple of 20 nanoseconds. Available suffixes are Ps (picoseconds), Ns (nanoseconds), Ms (milliseconds), and S (seconds). Default units are nanoseconds. Scientific notation is accepted. Negative numbers are not allowed.

Response: OK or ?? (to indicate an error).

Notes:

- Pulse widths are rounded to the nearest multiple of 20 nanoseconds when set. Use the AX PWidth command to read what value was applied.
- Pulse widths not equal to a multiple of 20 nanoseconds do not generate an error, but values out of the valid range do generate errors.
- Although discouraged, a pulse width of 0 is allowed and supported. In this case, pulses will never be observed.

Burst Commands

BURst

Description: Query the current burst configuration.

Response: Responds with BURst <enabled> N <n_pulses> of M <m_triggers>, where:

- <enabled> may be either ON or OFF, indicating the configured state of the burst functionality (set by the BURst ON and BURst OFF commands).
- <n_pulses> is the number of shots that will occur in a burst, akin to the 'numerator' of the burst logic. This is formatted like and equivalent to the response generated by the BNum command.
- <m_triggers> is the number of shots that compose a burst, akin to the 'denominator' of the burst logic. This is formatted like and equivalent to the response generated by the BMod command.

Notes:

- The response of this command is affected by verbose mode.
- If $N = 0$, $M = 0$, or $N > M$, burst functionality is disabled. Using bursts with $N = M$ is not advised since this may produce unintuitive behavior (bursts will be enabled if burst is configured as on).
- Even though burst may be reported as ON, the N and M configuration can force burst functionality disabled. This will not be reported by this command.
- Operating with the gate operation set to remote or gate burst mode with burst disabled (invalid N and M values satisfy this) behaves as though burst is enabled, with $N = 1$ and $M = 1$.

BUrst ON

Description: Enable burst mode.

Enables burst functionality.

Response: OK or ?? (to indicate an error).

Notes:

- Execution of this command clears *both* the predivider and burst counters.
- If $N = 0$, $M = 0$, or $N > M$, burst functionality is disabled. Using bursts with $N = M$ is not advised since this may produce unintuitive behavior (bursts will be enabled if burst is configured as on).

BUrst OFF

Description: Disable burst mode.

Disables burst functionality.

Response: OK or ?? (to indicate an error).

Notes:

- Execution of this command clears *both* the predivider and burst counters.

BUrst REset

Description: Reset the burst counter.

Immediately clears both the predivider and burst counters.

Response: OK or ?? (to indicate an error).

Notes:

- Execution of this command clears *both* the predivider and burst counters.

BNum

Description: Query the burst pulse count.

Response: Responds with the burst 'N' value as a 32-bit unsigned integer.

Notes:

- The response of this command is affected by verbose mode.

BNum <n_pulses>

Description: Set the burst pulse count.

Sets the burst 'N' value, and updates the burst configuration.

Parameters:

- <n_pulses> - The number of shots that will occur in a burst, akin to the 'numerator' of the burst logic.

Response: OK or ?? (to indicate an error).

Notes:

- Execution of this command clears *both* the predivider and burst counters.
- If $N = 0$, $M = 0$, or $N > M$, burst functionality is disabled. Using bursts with $N = M$ is not advised since this may produce unintuitive behavior (bursts will be enabled if burst is configured as on).
- Operating with the gate operation set to remote or gate burst mode with burst disabled (invalid N and M values satisfy this) behaves as though burst is enabled, with $N = 1$ and $M = 1$.

BMod

Description: Query the burst period count.

Response: Responds with the burst 'M' value as a 32-bit unsigned integer.

Notes:

- The response of this command is affected by verbose mode.

BMod <m_triggers>

Description: Set the burst period count.

Sets the burst 'M' value, and updates the burst configuration.

Parameters:

- <m_triggers> - The number of shots that compose a burst, akin to the 'denominator' of the burst logic.

Response: OK or ?? (to indicate an error).

Notes:

- Execution of this command clears *both* the predivider and burst counters.
- If $N = 0$, $M = 0$, or $N > M$, burst functionality is disabled. Using bursts with $N = M$ is not advised since this may produce unintuitive behavior (bursts will be enabled if burst is configured as on).
- Operating with the gate operation set to remote or gate burst mode with burst disabled (invalid N and M values satisfy this) behaves as though burst is enabled, with $N = 1$ and $M = 1$.

Clock Commands

Clock

Description: Query clock settings and board temperature.

Response: Responds with
Clock <mode> Trim <vcxo_trim> Temp <board_temp>, where:

- <mode> is OUT, if the clock connector is used as an output, PLL if used as an input, or HIZ if clock functionality is disabled.
- <trim> is the currently-configured VCXO trim, as a 16-bit unsigned integer, equivalent to the response of the CTrim command.
- <board_temp> indicates the current board temperature in degrees Celsius, formatted as +00.0 for positive values and -00.0 for negative values.

Clock HIZ

Description: Disable the clock connector.
Configure the clock connector to be disabled and undriven.

Response: OK or ?? (to indicate an error).

CLOCK OUT

Description: Output a 10 MHz clock via the clock connector.
Configure the clock connector to be utilized as an output. As an output, the clock signal has a 50 Ohm source termination and is driven at 3.3V TTL logic levels.

Response: OK or ?? (to indicate an error).

CLOCK IN

Description: Utilize the clock connector as an input and lock to an external 10 MHz clock.
Configure the clock connector to be utilized as an input, then attempt to lock to any present 10 MHz reference signal.

Response: OK or ?? (to indicate an error).

Notes:

- If the external 10 MHz clock is out of specification for frequency tolerance, the device may still succeed in locking but produce excessive output jitter. Although this behavior is unconfirmed, it is possible that this will not be visible as an error.
- Running this command will immediately assert the VCXO flag (see: ERRORS) since the VCXO requires time to lock. It is recommended to clear the error flags with the ERRORS <n> command until determining that the VCXO has locked successfully. This way, a loss-of-lock event will produce a noticeable error if it occurs in the future.
- While in clock input mode, the set VCXO trim (user- or factory-set) is ignored. If not locked to an external reference, the trim is undefined, and timing accuracy is not guaranteed.

CLOCK SAVE

Description: Save the user-set clock trim value to nonvolatile memory.
Save the user-set VCXO trim value to non-volatile memory, to be loaded at every power-up in place of the factory-calibrated trim value.

Response: OK or ?? (to indicate an error).

Notes:

- Currently, although this command does not overwrite the factory-calibrated trim value, there is no way to restore the VCXO trim to the factory-calibrated default value. Use with caution.

CTrim

Description: Query the VCXO trim value.
Reports the active VCXO trim value (not the value saved in non-volatile memory). If a user-mandated trim value has been set, this will override the factory-calibrated default trim value.

Response: The active trim value for the on-board oscillator, as a 16-bit unsigned integer. A value difference of 1 affects the oscillator frequency by approximately 7 ppt (parts per trillion), however this is not specified or guaranteed across units.

Notes:

- This command only reports the configuration status, not what the VCXO trim value is while locking to an external 10 MHz reference.

CTrim <trim>

Description: Set the VCXO trim value.

Parameters:

- <trim> - The active trim value for the on-board oscillator, as a 16-bit unsigned integer. A value difference of 1 affects the oscillator frequency by approximately 7 ppt (parts per trillion), however this is not specified or guaranteed across units.

Response: OK or ?? (to indicate an error).

Notes:

- This parameter requires the CLock SAve command to be loaded at power-up.
- Adjusting the VCXO trim too far from its initial value at power-up (further than the typical external reference locking range) can cause excessive output jitter. If this occurs, the user-configured trim value must be saved and the unit rebooted.
- Changes are applied immediately to hardware. If the clock is currently being used as an input, the effect of this command will not be visible until the clock mode is changed to output or disabled.

Trains Commands

TCount

Description: Query the number of pulses per trigger.

Reports the train count of the active configuration.

Response: Responds with the trains pulse count as an unsigned 32-bit unsigned integer. A value of 0 indicates that trains are disabled.

Notes:

- The response of this command is affected by verbose mode.

TCount Off

Description: Disable pulse trains.

Sets the trains pulse count to 0. Equivalent to executing the TCount <pulses> command with an argument of 0.

Response: OK or ?? (to indicate an error).

Notes:

- This command modifies the pending output configuration, and requires an autoinstall, install, or queue to take effect.

TCount <pulses>

Description: Set the number of pulses per trigger.

Sets the trains pulse count of the pending configuration. A value of 0 indicates that trains will be disabled.

Parameters:

- <pulses> - The number of *additional* pulses that will be generated during a shot, as a 32-bit unsigned integer. All 32-bit values are valid. A value of 0 indicates that trains are disabled.

Response: OK or ?? (to indicate an error).

Notes:

- This command modifies the pending output configuration, and requires an autoinstall, install, or queue to take effect.

TSpace

Description: Query the pulse spacing.

Reports the train spacing of the active configuration.

Response: Responds with the train spacing as an unsigned 32-bit unsigned integer, where a value of 1 represents 20 nanoseconds.

Notes:

- The response of this command is affected by verbose mode.

TSpace <time>

Description: Set the train pulse spacing.

Sets the trains pulse spacing of the pending configuration.

Parameters:

- <time> - The number of 20 nanosecond intervals between output pulses, as a 32-bit unsigned integer. Only values 3 and higher are allowed.

Response: OK or ?? (to indicate an error).

Notes:

- This command modifies the pending output configuration, and requires an autoinstall, install, or queue to take effect.

Frames Commands

FRame

Description: Report the frames engine status.

Reports the state of the frames engine, or the current frame pointer if it is running.

Response: Responds OFF if the frames engine is inactive, DONE if it is active but has completed running, or, if the frames engine is currently running, responds with the current frame pointer as a 16-bit unsigned integer (the same response generated by the FP command).

A response of ERROR is possible, indicating that the frames engine has been halted due to an internal error. Such a response is likely indicative of a hardware failure. In this state, the frames engine behaves like it is in a DONE state: active, but not running.

Notes:

- It is possible for this command to return ERROR, unlike the T560.

FRame OFF

Description: Cancel frame mode.

Deactivate the frames engine and force an install.

Response: OK or ?? (to indicate an error).

Notes:

- This command issues an install as though the INStall command were run, and carries all notes associated with that command.
- Running this command clears a pending autoinstall, regardless of autoinstall mode.
- This command terminates any possibly-running shot by issuing an end-of-delay (see: FEod).

FRame <frame>

Description: Store the current settings into the specified frame.
Saves the pending configuration to the specified frame.

Parameters:

- <frame> - The index of the frame to save to.

Response: OK or ?? (to indicate an error).

Notes:

- In order for this command to be operable, the frames engine must be inactive (in the DONE or OFF states). If this condition is not met, this command will produce a ?07: Illegal operation error.

FRame GO

Description: Start frames operation.

Activates the frames engine, then starts it. If the frames configuration is valid, the frames engine will then be moved into the running state and the first frame will immediately be loaded into hardware.

The frames configuration is considered invalid if the index of the last frame to be executed (FB) is not greater than the first frame index (FA). In this case, this command will produce a ?08: Settings conflict error and the frames engine will not be started.

Response: OK or ?? (to indicate an error).

Notes:

- This command terminates any possibly-running shot by issuing an end-of-delay (see: FEod).
- To preserve T560 compatibility, running this command increments the loaded frame count by 2, instead of the expected 1.

FRame LAsT

Description: Report the last available frame index.

Response: Responds with the maximum number of frames supported by the device as a 16-bit unsigned integer. This is defined to always be 08191.

FDump

Description: Display the current frame's contents.

Prints the contents of frame 0's timing configuration, in an internal diagnostics format. Although the response of this command is designed to match that of the T560, the T660's frame configuration contains more data than is represented by the response of this command. Therefore, this command is provided for compatibility reasons and matches the T560's behavior as closely as possible.

Unintuitively, running this command is functionally equivalent to running the FDump <frame> command with an argument of 0.

Response: Begins with a line delimiter, followed by ten lines (the last without a trailing line delimiter) consisting of frame data. The specific meaning of the returned values follows an internal format; while the returned values behave similarly to the T560, they are not exactly identical in meaning.

Notes:

- This command is deprecated, and is provided solely for T560 compatibility.
- This command produces a response that is multiple lines long (the response includes line-endings).
- Note that this command does not report the current or starting frame's contents, but those of frame 0.

FDump <frame>

Description: Display a frame's contents.

Prints the contents of a frame's timing configuration, in an internal diagnostics format. Although the response of this command is designed to match that of the T560, the T660's frame configuration contains more data than is represented by the response of this command. Therefore, this command is provided for compatibility reasons and matches the T560's behavior as closely as possible.

Parameters:

- <frame> - The index of the frame to inspect the contents of.

Response: Begins with a line delimiter, followed by ten lines (the last without a trailing line delimiter) consisting of frame data. The specific meaning of the returned values follows an internal format; while the returned values behave similarly to the T560, they are not exactly identical in meaning.

Notes:

- This command is deprecated, and is provided solely for T560 compatibility.
- This command produces a response that is multiple lines long (the response includes line-endings).

FA

Description: Query the start frame index.

Response: The index of the first frame to be executed, as a 16-bit unsigned integer.

FA <frame>

Description: Set the start frame index.

Parameters:

- <frame> - The index of the first frame to be executed.

Response: OK or ?? (to indicate an error).

Notes:

- In order for this command to be operable, the frames engine must be inactive (in the DONE or OFF states). If this condition is not met, this command will produce a ?07: Illegal operation error.

FB

Description: Query the end frame index.

Response: The index of the last frame to be executed, as a 16-bit unsigned integer.

FB <frame>

Description: Set the end frame index.

Parameters:

- <frame> - The index of the last frame to be executed.

Response: OK or ?? (to indicate an error).

Notes:

- In order for this command to be operable, the frames engine must be inactive (in the DONE or OFF states). If this condition is not met, this command will produce a ?07: Illegal operation error.

FC

Description: Query the frames loop count.

Response: Responds with the number of loops the frames engine shall perform, as a 16-bit unsigned integer. A value of 0 means that only one iteration will be performed, 1 means that two iterations will be performed, etc. A value of 65535 (or 0xFFFF) represents infinity, and the frames engine will never cease operation.

FC <count>

Description: Set the frames loop count.

Sets the number of loops the frames engine shall perform, with the starting and ending frames determined by the FA and FB commands, respectfully.

Parameters:

- <count> - The number of loops the frames engine shall perform, as a 16-bit unsigned integer. A value of 0 means that only one iteration will be performed, 1 means that two iterations will be performed, etc. A value of 65535 (or 0xFFFF) represents infinity, and the frames engine will never cease operation.

Response: OK or ?? (to indicate an error).

Notes:

- In order for this command to be operable, the frames engine must be inactive (in the DONE or OFF states). If this condition is not met, this command will produce a ?07: Illegal operation error.

FN

Description: Query the number of frames loaded into hardware.

Reports the number of frames loaded into hardware by the frames engine.

Response: The number of frames loaded while the frames engine has been active, since last power-on or FN <count> command execution.

FN <count>

Description: Clear the loaded frame count.

Clears the loaded frame counter. This command may be run while the frames engine is active and running.

Parameters:

- <n> - May be any integral number. This number is ignored.

Response: OK or ?? (to indicate an error).

FM

Description: FM

Response: A decimal, 32-bit integer representing the number of frame loads remaining before the frame engine is in a DONE state. A value of 0xFFFFFFFF indicates that the engine is either currently in a DONE state or the remaining loop count represents infinity.

Notes:

- The value returned by this command should be considered undefined if the frames engine is inactive.

FP

Description: Query the current frame pointer.

Reports the index of the currently-loaded frame in hardware. The next valid incoming trigger will use this frame's configuration to produce outputs.

Response: Responds with index the currently-active frame as a 16-bit unsigned integer.

Notes:

- The value of the frame pointer is undefined while the frames engine is inactive. Although the behavior matches the T560's, using this value while the frames engine is inactive is not recommended.

FX

Description: Report frames engine diagnostics.

Prints the current state of the frames engine, in an internal format. Although the response of this command is designed to match that of the T560, the T660's frame engine state contains more data than is represented by the response of this command. Therefore, this command is provided for compatibility reasons and matches the T560's behavior as closely as possible.

with a line delimiter then two lines, the second with no trailing line delimiter. The first line contains the following fields: FC (frames loop count), FAIM (starting frame index), FSTATE (frames engine state), FOX (frames engine off code), FREKLE (frames engine error code), and FLOAD (frames engine load count). The second line contains the following fields: <end_index> (ending frame index), <loop_count> (frames loop count), <pointer> (current frame pointer), <state> (current frames engine state), <remaining_msbs> (upper 16 bits of remaining frame loads), <remaining_lsbs> (lower 16 bits of remaining frame loads), <load_count_msbs> (upper 16 bits of frame load counter), and <load_count_lsbs> (lower 16 bits of frame load counter).

<end_index> is the starting frame index, the same as reported by the FA command.

<index> is the ending frame index, the same as reported by the FB command.

<count> is the frames loop count, the same as reported by the FC command.

<er> is the current frame pointer, the same as reported by the FP command.

<state> is the current state of the frames engine, 0x0000 for off, 0x0001 for running, 0xFFFF for done, and 0x8000 for error. Other values are reserved for future use and may indicate error codes.

<remaining_msbs> and <remaining_lsbs> compose the upper 16 and lower 16 bits of the 32-bit number of frame loads remaining before the engine reaches a DONE state. 0xFFFF indicates that the engine is either in a DONE state or the loop count represents infinity.

<count_msbs> and <load_count_lsbs> compose the upper 16 and lower 16 bits of the 32-bit frame load counter.

Notes:

- This command is deprecated, and is provided solely for T560 compatibility.
- This command produces a response that is multiple lines long (the response includes line-endings).

RTest

Description: Test the frames memory.

Performs a readback test of the frames memory to ensure its proper operation.

Response: OK or ?? (to indicate an error).

Notes:

- This command is deprecated, and is provided solely for T560 compatibility.

RZap

Description: Clear the frames memory.

Write the default configuration to all frames.

Response: OK or ?? (to indicate an error).

Notes:

- In order for this command to be operable, the frames engine must be inactive (in the DONE or OFF states). If this condition is not met, this command will produce a ?07: Illegal operation error.
- Unlike the T560, which clears the frames memory to an undefined state, this command sets every frame to the default configuration's timing settings.
- This command is not allowed to be run while the frames engine is active (not to be confused with running). This is not the case in the T560.

Memory Commands

SAve

Description: Save the current configuration to nonvolatile memory.

All settable configuration is saved to non-volatile memory immediately. Exceptions to this are the oscillator configuration (VCXO trim), frames memory, network configuration, and serial configuration (serial baud rate). These exceptions are saved and loaded independently from the memory commands.

Response: OK or ?? (to indicate an error).

Notes:

- Unlike other memory commands, running this command does not impact device operation.
- Frames memory is not saved to nonvolatile memory.
- Clock configuration is saved by this command, however VCXO trim is not.

REcall

Description: Load the system configuration from nonvolatile memory.

Loading the stored configuration will ensure that all configuration is re-applied, ensuring that no prior commands have affected the unit's state. Exceptions to this are the oscillator configuration (VCXO trim), frames memory, network configuration, and serial configuration (serial baud rate). These exceptions are saved and loaded independently from the memory commands.

Completion of this command may result in immediate activity by the unit, since whatever configuration was saved is immediately applied.

Response: OK or ?? (to indicate an error).

Notes:

- This command modifies all configuration settings except active network and clock configuration.
- Frames memory is not affected by this command.
- Running this command stops frames engine operation, equivalently to running FFrame OFF.
- This command terminates any possibly-running shot by issuing an end-of-delay (see: FEod).
- Running this command clears a pending autoinstall, regardless of autoinstall mode.

LOad DEfault

Description: Load configuration defaults.

Loading the default configuration will ensure that all configuration is re-applied, ensuring that no prior commands have affected the unit's state. Exceptions to this are the oscillator configuration (VCXO trim), frames memory, network configuration, and serial configuration (serial baud rate). These exceptions are saved and loaded independently from the memory commands.

Completion of this command will result in the unit becoming idle due to the trigger source being set to remote.

Response: OK or ?? (to indicate an error).

Notes:

- This command modifies all configuration settings except active network and clock configuration.
- Frames memory is not affected by this command.
- Running this command stops frames engine operation, equivalently to running FRame OFF.
- This command terminates any possibly-running shot by issuing an end-of-delay (see: FEod).
- Running this command clears a pending autoinstall, regardless of autoinstall mode.

RUn DEmo

Description: Load the demonstration configuration.

The demonstration configuration differs from the default configuration in that:

- The trigger source is set to internal (as opposed to remote)
- The predivider divisor is set to 4000 (as opposed to 0)

Completion of this command will result in the unit outputting pulses immediately due to this configuration.

Response: OK or ?? (to indicate an error).

Notes:

- This command modifies all configuration settings except active network and clock configuration.
- Frames memory is not affected by this command.
- Running this command stops frames engine operation, equivalently to running FRame OFF.
- This command terminates any possibly-running shot by issuing an end-of-delay (see: FEod).
- Running this command clears a pending autoinstall, regardless of autoinstall mode.

Network Commands

SBaud

Description: Query the serial port baud rate.

Response: Responds with the current operating RS232 serial port baud rate, formatted as an integer. This will be one of: 9600, 19200, 38400, 57600, 115200, 230400, or 460800.

SBaud <baud>

Description: Set the serial port baud rate.

Configures the baud rate of the RS232 serial port, then saves this baud rate to non-volatile memory. The configured baud rate will also be applied at device power-on until reconfigured.

Parameters:

- <baud> - The baud rate, as an integer. Valid values are 9600, 19200, 38400, 57600, 115200, 230400, or 460800.

Response: OK or ?? (to indicate an error).

Notes:

- Take caution when running this command over the serial interface: the baud rate changes during command execution. Because of this, the OK response by this command will be output in the newly-configured baud rate.
- Running this command with a <baud> parameter equal to the current baud rate will not interrupt serial communications.

NEtwork

Description: Report the network status and active configuration.

Response: Responds with the current network status (not to be confused with the active network configuration) in the form
Net <status> Addr <address> Mask <subnet> Status <connected>,
where:

- <status> may be STATIC, DHCP, or OFF, depending on the active network configuration. A reported status of OFF may indicate a hardware failure if the device features hardware Ethernet connectivity.
- <address> is the static IPv4 address or assigned DHCP IPv4 address in the format X.X.X.X, or PENDING if awaiting DHCP assignment.
- <subnet> is the static IPv4 subnet mask or assigned DHCP IPv4 subnet mask in the format X.X.X.X, or PENDING if awaiting DHCP assignment.
- <connected> is either UP or DOWN, representing the if the Ethernet connector is currently connected. Note that this status represents the physical connection status, not the network connection status.

NEtwork Address <ipv4_address>

Description: Set the IPv4 address for the pending network configuration.

Sets the pending network configuration's IPv4 address.

Parameters:

- <address> - The IPv4 address, in the form X.X.X.X. 0.0.0.0 indicates that DHCP shall be used to obtain an IPv4 address.

Response: OK or ?? (to indicate an error).

Notes:

- Invalid IP addresses other than 0.0.0.0 (such as those not matching the set IPv4 subnet mask) may produce undefined behavior and are not supported.

NEtwork Address

Description: Query the IPv4 address set by the active network configuration.

Query the pending IPv4 address configuration. Note that since this represents the pending configuration, not the current network status, a valid response is 0.0.0.0, indicating that DHCP is to be configured. At power-up, this is 0.0.0.0.

Response: The most recently applied IPv4 address configuration, formatted as X.X.X.X.

Network Apply

Description: Apply and save the pending network configuration.

Applies the pending network configuration to the active network configuration. This will also immediately terminate any network connections to the device while the Ethernet connection is restarted. Once applied, the network configuration will be saved to non-volatile memory.

Response: OK or ?? (to indicate an error).

Notes:

- Running this command from network sessions will terminate the session from which the command was issued; there is no guarantee that a response will be sent back over the network interface (it likely will not).
- If the network configuration is set to DHCP using the Network Address command, the value set by the Network Subnet <ipv4_subnet> command is effectively ignored, and may not be saved to non-volatile memory.
- Running this command immediately after power-on will set the network configuration to DHCP, due to the power-on default values of the pending network configuration.

Network Hostname

Description: Query the device hostname.

Response: Responds with the device's serial number in the format T660-XXXXX, where XXXXX is the serial number. This returned string is the name with which the device may be accessed using DHCP and uDNS.

Network MAC

Description: Query the device MAC address.

Response: Responds with the device's Ethernet MAC address in the form XX:XX:XX:XX:XX:XX.

Network Subnet <ipv4_subnet>

Description: Set the IPv4 subnet for the pending network configuration.

Parameters:

- <ipv4_subnet> - The IPv4 subnet mask, in the form X.X.X.X.

Response: OK or ?? (to indicate an error).

Notes:

- Invalid subnet masks (such as those not matching the set IPv4 address) may produce undefined behavior and are not supported.
- If the network configuration is set to DHCP using the Network Address command, this value is effectively ignored once applied, and may not be saved to non-volatile memory.

Network Subnet

Description: Query the IPv4 subnet set by the active network configuration.

Response: The pending IPv4 subnet configuration, formatted as X.X.X.X. At power-up, this is 255.255.255.0.

Notes:

- If the network configuration is set to DHCP as reported by the `NEtwork ADdress` command, the reported value may be the default value.

Help Commands

Since firmware version 1.3, all T560-style help commands have been deprecated. To read command documentation, refer to this programming guide. For T560 compatibility, the help commands have been kept, but now respond with a deprecation warning when queried.

P500-Style Syntax

Basic Syntax

P500-style syntax is a modified SCPI syntax. Each command or query consists of a phrase consisting of one or more keywords, separated by colons (:) as necessary. At the end of the command phrase, an optional question mark (?) indicates that the command should be interpreted as a query. Most commands can also be executed as queries. Arguments follow commands and queries; the first argument is separated from the command phrase by whitespace, and subsequent arguments by commas (,).

For example, in the command `CHAN:ACT:MOD A, OFF`, `CHAN:ACT:MOD` is the command phrase and `A` and `OFF` the arguments. This would turn off channel A output pulsing. `CHAN:ACT:MOD? A` would query the enable state of the channel.

Just like in T560-style syntax, multiple commands may be sent in a single *command line*, separated by semicolons. Unlike T560-style syntax, colons (:) may not be used to delimit commands due to their usage in command phrases. When a full line is received, indicated by a line delimiter (`<cr>`, `<lf>`, or `<cr><lf>`), the buffered line's commands are executed in the order received.

P500-style command lines do not accept “empty” commands. A command line consisting of `;` would produce a `?21: Command name expected error`. Lines consisting entirely of whitespace, such as `<space><cr><lf>` will produce the same result, however entirely empty lines consisting of a line delimiter only (`<cr>`, `<lf>`, or `<cr><lf>`) will respond with an empty line (`<cr><lf>`).

Each command generates a *response*, and each command line generates a *response line*, consisting of all the commands' responses in the order executed, separated by semicolon (;) characters. Every response line is terminated with the carriage return and line-feed characters (`<cr><lf>`).

Multiple Commands Per Line

After the first command of a command line, additional commands on the same line use the context of the previously-executed command to infer leading token keywords. This means that unintuitively, the command line `CHAN:MOD? A; CHAN:MOD? B` will produce a `?22: Command not recognized error`.

The first command that executes sets the subcommand tokens: `CHAN:MOD? A` refers to the `CHANnel[:ACTive]:MODE? <channel>` command, which sets the subcommand phrase to `CHANnel:ACTive:` for the following commands. The next command is interpreted as though this phrase preceded it. So, in this example, the command line `CHAN:MOD? A; MOD? B` would be correctly formed.

To avoid using the previous command's subcommand phrase for a given command, precede the command phrase with a colon (:). Thus, `CHAN:MOD? A, :GATE:MOD?` is a valid command line, but `CHAN:MOD? A, GATE:MOD?` (no colon preceding the gate command) is not. An additional command would be executed with the `GATE:` subcommand phrase assumed, unless preceded by a colon.

Command Structure

Every command, query, and subcommand keyword has as long and short form, both of which are recognized equivalently. These are denoted with uppercase and lowercase letters in this documentation, for short and long, respectively. Forms may be mixed within a single command phrase.

For example, the `GATE:INPut:TERMination?` query can be executed one of four ways:

- `GATE:INP:TERM?`
- `GATE:INPUT:TERM?`
- `GATE:INP:TERMINATION?`
- `GATE:INPUT:TERMINATION?`

Sometimes, command keywords may be omitted. In these cases, this documentation will surround the optional tokens with brackets ([and]). For example, the `CHANnel[:ACTive]:MODE? <channel>` command could be called as `CHANnel:ACTive:MODE?` or `CHANnel:MODE?`.

Some keywords serve only to represent subcommands. In these cases, attempting to run the keyword as a command or query will produce an error. For example, attempting to run CHANnel? would produce a ?23: Subcommand expected error.

Response Structure

In P500-style sessions, each command or query is *guaranteed* to only produce a single-line response, and without additional space characters. This means that each command line is also guaranteed to only produce one line delimiter as a consequence of its execution.

Both commands and queries may execute complicated or time intensive operations, if necessary, and are only differentiated by their responses. All commands (not queries) produce the response OK, or produce an error. All queries reply with relevant data (never OK), or produce an error.

Although nearly every command only responds with a single piece of data, when necessary, multiple response data are separated by commas (,), without additional whitespace. For example, SYSTem:TEMPerature:ALL? might respond with 61.520,43.500.

Time Formatting

A *time* is a special data type for P500-style commands. A time is a high-precision, integereal data type with a minimum quantity of 1 picosecond, expressed in seconds.

As a Prameter

P500, T560, and SCPI formatting are all accepted numerical formatting methods, and all allow for full precision. It is suggested to use a format similar to the response a P500 gives, although this is not required. Scientific notation is allowed, with an optional +, -, or single space character before the exponent to indicate exponent polarity. All characters are case-insensitive.

Allowable suffixes are: Ps (picoseconds), Ns (nanoseconds), Us (microseconds), Ms (milliseconds), S (seconds). The default suffix is seconds.

For example, the text 1.2000E 9p, 1.2ms, +1200e-6, and +00.0012000 are all valid representations of 1.2 milliseconds.

As a Response

Times are formatted as +000.000000000000, a number representing seconds. A negative time is preceded by a - instead of a +. Times greater than or equal to an absolute value of 1000 seconds are expressed with additional digits, like +1000.000000000000 or -1000.000000000000.

Configuring Pulse Outputs

Timing Edges

In P500 sessions, the delay and width of channel outputs are configured on a timing edge basis, rather than per channel. Edge indices are used to specify edges throughout the P500-style timing commands.

There are 9 timing edges, 8 of which are configurable. All configurable edges correspond to the rising and falling edges of channel outputs and are called channel edges. A list of all edges and their associated index is below.

Index	Description
0	Start of shot, at minimum insertion delay.
1	Channel A rising edge.
2	Channel A falling edge.
3	Channel B rising edge.
4	Channel B falling edge.
5	Channel C rising edge.
6	Channel C falling edge.
7	Channel D rising edge.
8	Channel D falling edge.

Note: For all commands and queries, edge indices are considered strings, not numbers, so 01 is not a valid representation of edge 1.

Channel Timing Mode

Output channels (A, B, C and D) have a configuration property called timing mode. They may be configured to either delay-width mode or rise-fall mode. This configuration property is global and unaffected by installs, and serves only to modify the behavior of the P500-Style [Timing Commands](#).

In rise-fall mode, a channel's rising and falling edges are completely decoupled from one another, except for the requirement that the falling edge occurs after the rising edge in absolute time. When the timing of an edge is modified, edges relative to it are unaffected (see [Edge Relativity](#)) with respect to absolute delay.

In delay-width mode, a channel's falling edge is forced to be relative to the channel's rising edge. Whenever the timing of the rising edge is updated, the falling edge is updated, too. Note, however, that rising edge [Edge Relativity](#) behavior still behaves identically to rise-fall mode, and other dependent edges referenced to one being reconfigured are not updated in absolute time.

Additionally, note that running T560-style timing commands (any command that either sets or queries any timing configuration) forces a channel's timing mode to be delay-width and its rising edge to be relative to edge 0 (shot start) when any such command is run.

Edge Relativity

Channel edges may be configured for their delays to be expressed relative to another timing edge by setting a reference edge using the `TIME:RELTo<edge> <reference_edge>` command. When an edge is configured to be relative to another edge, all time configuration command parameters should be expressed relative to the configured relative edge, and all queries will result in responses relative to the relative edge.

Below is an example configuration where channels A, B, C, and D are set to delays of 10, 30, 50, and 70ns each, all with a width of 20ns. The actual delay, relative to edge index 0 (the start of a shot), represents the absolute time between the start of a shot and the rising/falling edge of a channel's output. The reference edge and relative delay are the configuration of the specified channel edge, as observed through commands.

Edge	Actual Delay (ns)	Reference Edge	Relative Delay (ns)
1	10	0	10
2	20	1	10
3	30	0	30
4	40	2	20
5	50	7	-20
6	60	8	-20
7	70	4	30
8	80	0	80

Edge relativity follows the following notable rules:

- An edge may not be relative to itself.
- A channel's rising edge must occur after its falling edge in absolute time.
- When the relativity for an edge is modified, its absolute delay is retained; only commands and queries to the configuration are affected.
- Edge relativity is a global configuration per edge that applies to all timing sets (pending, active, and queued) immediately.
- When an edge delay is set, dependent edge timings are *not* updated relative to shot start (unless the edge is the rising edge of a channel in delay-width mode, in which case the falling edge for that channel is also updated).
- Edge relativity is affected by [Channel Timing Mode](#): if a channel is in delay-width mode, all falling edges are forced to be relative to their channel's rising edge and are updated automatically with rising edge changes.
- When setting a delay, absolute delay is calculated relative to the pending timing configuration.
- When querying a delay, delays are calculated relative to the timing configuration specified. This means queued delays will be calculated relative to other queued delays.
- All relative delays are calculated at the moment a command executes, and are thus subject to interference by multiple command sessions.
- While circular references are technically allowed, they are highly discouraged since they behave unintuitively.

Additionally, note that running T560-style Timing Commands (any command that either sets or queries a channel's delay or width) or the SStatus (all channels affected) command forces a channel's timing mode to be delay-width and its rising edge to be relative to edge 0 (shot start) when any such command is run. This is important to keep in mind when using T560-style and P500-style sessions simultaneously.

P500-Style Commands List

Command Table

Below is a list of P500-style commands that can be executed while operating the T660 through a P500-style session. Not all commands are available to all T660 models; some commands or operations require the T660 to be purchased with the *Trains and Frames* feature.

Command	Mode Is	Brief Description
Global Commands		
*IDN?	All	Identify the device.
*RST	All	Reboot the device.
STArT	All	Allow triggers and resume pulse generation functionality.
STOp	All	Stop outputting pulses and responding to triggers.
INSTAlL	All	Apply pending settings immediately.
QUEue	T&F	Queue settings to be installed at next EOD.
fEOD	All	Force an end-of-delay to occur.
Session Commands		
ZMode <mode>	All	Set the current command session type.
ZMode?	All	Query the current command session type.
SESSion:WAIT <us>	All	Wait a specified number of microseconds.
SESSion:VERBosity[:ERRor] <level>	All	Set the session error verbosity level.
SESSion:VERBosity[:ERRor]?	All	Query the session error verbosity level.
System Status Commands		
SYSTem:ERRor:FLAGs <n>	All	Clear global error flags.
SYSTem:ERRor:FLAGs?	All	Query global error flags.
SYSTem:BIST:POWer?	All	Query the comprehensive device power supply status.
SYSTem:BIST:STATus?	All	Query the background BIST status.
SYSTem:TEMPerature[:ALL]?	All	Query the CPU and PCB temperatures, in degrees Celsius.
SYSTem:TEMPerature:BOARD?	All	Query the PCB temperature, in degrees Celsius.
SYSTem:TEMPerature:CHIP?	All	Query the CPU temperature, in degrees Celsius.
SYSTem:MICROSeconds <n>	All	Clear the microsecond counter.
SYSTem:MICROSeconds?	All	Query the microsecond counter.
FEATure:FRAME?	All ⁴	Query whether the frames feature is available on the device.
FEATure:STATus:FRAME?	All ⁴	Query whether the frames feature is available on the device.

Command	Mode Is	Brief Description
Channel Commands		
CHANnel:COMmit	All	Apply pending timing settings immediately.
CHANnel:SCHedule	T&F	Queue all pending timing settings.
CHANnel[:ACTive]:MODE <channel>, <enable>	All	Enable or disable a channel's output.
CHANnel[:ACTive]:MODE? <channel>	All	Query the output enable state of a channel.
CHANnel[:ACTive]:POLarity <channel>, <polarity>	All	Configure a channel's output polarity.
CHANnel[:ACTive]:POLarity? <channel>	All	Query a channel's output polarity.
CHANnel[:ACTive]:TERMination <channel>, <termination>	All	Configure a channel's output termination.
CHANnel[:ACTive]:TERMination? <channel>	All	Query a channel's output termination.
CHANnel:NEXT:MODE <channel>, <enable>	T&F	Queue a channel output enable state.
CHANnel:NEXT:MODE? <channel>	T&F	Query a channel's queued output enable state.
CHANnel:NEXT:POLarity <channel>, <polarity>	T&F	Queue a channel output polarity.
CHANnel:NEXT:POLarity? <channel>	T&F	Query a channel's queued output polarity.
CHANnel:NEXT:TERMination <channel>, <termination>	T&F	Queue a channel output termination.
CHANnel:NEXT:TERMination? <channel>	T&F	Query a channel's queued output termination.
CHANnel:QUEue:MODE <channel>, <enable>	All	Set a channel's pending output enable state.
CHANnel:QUEue:MODE? <channel>	All	Query a channel's pending output enable state.
CHANnel:QUEue:POLarity <channel>, <polarity>	All	Set a channel's pending output polarity.
CHANnel:QUEue:POLarity? <channel>	All	Query a channel's pending output polarity.
CHANnel:QUEue:TERMination <channel>, <termination>	All	Set a channel's pending output termination.
CHANnel:QUEue:TERMination? <channel>	All	Query a channel's pending output termination.
CHANnel:TimingMODE <channel>, <timing_mode>	All	Set a channel's timing mode.
CHANnel:TimingMODE? <channel>	All	Query a channel's timing mode.
CHANnel:ON <channel>	All	Enable a channel's output.
CHANnel:ON? <channel>	All	Query the output enable state of a channel.
CHANnel:OFF <channel>	All	Disable a channel's output.

Command	Mode Is	Brief Description
CHANnel:OFF? <channel>	All	Query the output enable state of a channel.
CHANnel:POSitive <channel>	All	Set a channel's output polarity to be active-high (positive)
CHANnel:POSitive? <channel>	All	Query the output polarity of a channel.
CHANnel:NEGative <channel>	All	Set a channel's output polarity to be active-low (negative)
CHANnel:NEGative? <channel>	All	Query the output polarity of a channel.
CHANnel:DelayWidth <channel>	All	Set a channel's timing mode to be delay-width.
CHANnel:DelayWidth? <channel>	All	Query the timing mode of a channel.
CHANnel:RiseFall <channel>	All	Set a channel's timing mode to be rise-fall.
CHANnel:RiseFall? <channel>	All	Query the timing mode of a channel.
CHANnel:500HM <channel>	All	Enable a channel's output termination.
CHANnel:500HM? <channel>	All	Query a channel's output termination.
CHANnel:LOWZ <channel>	All	Disable a channel's output termination.
CHANnel:LOWZ? <channel>	All	Query a channel's output termination.
Timing Commands		
TIME:COMmit	All	Apply pending timing settings immediately.
TIME:SCHedule	T&F	Queue all pending timing settings.
TIME:DELay<edge> <delay>	All	Set and install timing edge's delay.
TIME:DELay<edge>?	All	Query a timing edge's active delay.
TIME:INSDel <mode>	All	Set the insertion delay mode.
TIME:INSDel?	All	Query the insertion delay mode.
TIME:NEXT<edge>	All	Set and queue a timing edge delay change.
TIME:NEXT<edge>?	All	Query a timing edge's queued delay.
TIME:QUEue<edge> <delay>	All	Set a timing edge's pending delay.
TIME:QUEue<edge>?	All	Query a timing edge's pending delay.
TIME:RELTo<edge> <reference_edge>	All	Set a timing edge's reference edge.
TIME:RELTo<edge>?	All	Query a timing edge's reference edge.
TIME:UNDO	All	Revert changes to the pending configuration.
Trigger Commands		
TRIGger:EXECute	All	Fire a remote trigger.
TRIGger[:EXTernal]:PREDiv <divisor>	All	Query the trigger pre-divider divisor value.
TRIGger[:EXTernal]:PREDiv?	All	Set the trigger pre-divider divisor value.
TRIGger:FREQuency[:SYNth] <frequency>	All	Set the frequency of the on-board trigger synthesizer.
TRIGger:FREQuency[:SYNth]?	All	Query the frequency of the on-board trigger synthesizer.

Command	Mode Is	Brief Description
TRIGger:FREQuency:HITs?	All	Query the hit frequency counter.
TRIGger:FREQuency:TRIGger?	All	Query the trigger frequency counter.
TRIGger:PERiod[:SYNth]?	All	Query the period of the on-board trigger synthesizer.
TRIGger:PERiod:HITs?	All	Query the hit period counter.
TRIGger:PERiod:TRIGger?	All	Query the trigger period counter.
TRIGger:SOURce <source>	All	Set the trigger source.
TRIGger:SOURce?	All	Query the current trigger source.
TRIGger:SHOTs <n>	All	Clear the shot counter.
TRIGger:SHOTs?	All	Query the shot counter value.
TRIGger:INPut:POLarity <polarity>	All	Set the external trigger polarity.
TRIGger:INPut:POLarity?	All	Query the external trigger polarity.
TRIGger:INPut:TERMination <termination>	All	Enable or disable trigger connector end-termination.
TRIGger:INPut:TERMination?	All	Query the trigger connector end-termination state.
TRIGger:INPut:VOLTage <voltage>	All	Query the external trigger voltage threshold.
TRIGger:INPut:VOLTage?	All	Set the external trigger voltage threshold.
Gate Commands		
GATE:MODE <mode>	All ⁵	Set the gate operating mode.
GATE:MODE?	All	Query the current gate operating mode.
GATE:INPut:TERMination <termination>	All	Enable or disable gate connector end-termination.
GATE:INPut:TERMination?	All	Query the gate connector end-termination state.
GATE:EXECute	All	Fire a single remote burst.
Auxiliary Output Commands		
SYSTem:AUXiliary:MODE <mode>	T&F	Configure the auxiliary output operating mode.
SYSTem:AUXiliary:MODE?	T&F	Query the auxiliary output operating mode.
SYSTem:AUXiliary:UPDate:WIDTH <time>	T&F	Set the auxiliary output update mode pulse width.
SYSTem:AUXiliary:UPDate:WIDTH?	T&F	Query the auxiliary output update mode pulse width.
Burst Commands		
BURst:CClear	All	Clear the burst counter.
BURst:CClear?	All	Query the current burst counter count.
BURst:MODE <enable>	All	Enable or disable burst mode.
BURst:MODE?	All	Query whether burst mode is enabled.
BURst:PULse <n_pulses>	All	Set the burst pulse count.

Command	Mode Is	Brief Description
BURst:PULse?	All	Query the burst pulse count.
BURst:TRIGger <m_triggers>	All	Set the burst period count.
BURst:TRIGger?	All	Query the burst period count.
Clock Commands		
CLOCK:MODE <mode>	All	Set the clock connector operating mode.
CLOCK:MODE?	All	Query the clock connector operating mode.
CLOCK:EXTernal <enable>	All	Enable or disable locking to an external clock.
CLOCK:EXTernal?	All	Query whether external clock locking is enabled.
CLOCK:FREquency?	All	Query the perceived external clock frequency.
CLOCK:OCX0?	All	Query whether an OCXO is installed.
CLOCK:STATus?	All	Query the status of locking to an external clock.
CLOCK:TRIM[:SET] <trim>	All	Set the VCXO trim value.
CLOCK:TRIM[:SET]?	All	Query the effective VCXO trim value.
CLOCK:TRIM:DEFault	All	Restore the factory default VCXO trim value.
CLOCK:TRIM:SAVE	All	Save the user-set VCXO trim value to nonvolatile memory.
Trains Commands		
TRAI n[:ACTive]:CouNT <pulses>	T&F	Set the number of train pulses per trigger.
TRAI n[:ACTive]:CouNT?	T&F	Query the number of train pulses per trigger.
TRAI n[:ACTive]:SPACe <time>	T&F	Set train pulse spacing.
TRAI n[:ACTive]:SPACe?	T&F	Query train pulse spacing.
TRAI n:NEXT:CouNT <pulses>	T&F	Set the queued train count.
TRAI n:NEXT:CouNT?	T&F	Query the queued train count.
TRAI n:NEXT:SPACe <time>	T&F	Set the queued train spacing.
TRAI n:NEXT:SPACe?	T&F	Query the queued train spacing.
TRAI n:QUEue:CouNT <pulses>	T&F	Set the pending train count.
TRAI n:QUEue:CouNT?	T&F	Query the pending train count.
TRAI n:QUEue:SPACe <time>	T&F	Set the pending train spacing.
TRAI n:QUEue:SPACe?	T&F	Query the pending train spacing.
Frames Commands		
TFRame:STATus?	T&F	Report the frames engine status.
TFRame:PoinTeR?	T&F	Query the current frame pointer.
TFRame:REMaining?	T&F	Query the number of frames remaining to be loaded.
TFRame:LoadCouNT <n>	T&F	Clear the loaded frame count.

Command	Mode Is	Brief Description
TFRame:LoadCouNT?	T&F	Query the number of frames loaded into hardware.
TFRame:STArT	T&F	Start frames operation.
TFRame:STOp	T&F	Cancel frame mode.
TFRame:INSTALL <frame>	T&F	Apply a frame's timing settings immediately.
TFRame:QUEue <frame>	T&F	Queue a frame's timing settings to be installed at next EOD.
TFRame:STORe <frame>	T&F	Store the pending configuration into the specified frame.
TFRame:LOOP:FIRST <frame>	T&F	Set the start frame index.
TFRame:LOOP:FIRST?	T&F	Query the start frame index.
TFRame:LOOP:LAST <frame>	T&F	Set the end frame index.
TFRame:LOOP:LAST?	T&F	Query the end frame index.
TFRame:LOOP:CoUNT <count>	T&F	Set the frames loop count.
TFRame:LOOP:CoUNT?	T&F	Query the frames loop count.
TFRame:MEMOry:CLEAr	T&F	Clear the frames memory.
Memory Commands		
MEMOry:LOAD[:NORMAl]	All	Load the system configuration from nonvolatile memory.
MEMOry:LOAD:DEFault	All	Load configuration defaults.
MEMOry:LOAD:DEMO	All	Load the demonstration configuration.
MEMOry:SAVE[:ACTive]	All	Save the current configuration to nonvolatile memory.
Network Commands		
SYSTem:COMMUnicate:SERial:BAUD <baud>	All	Set the serial port baud rate.
SYSTem:COMMUnicate:SERial:BAUD?	All	Query the serial port baud rate.
SYSTem:COMMUnicate:SOCKeT:ADDReSS <address>	All	Set the IPv4 address for the pending network configuration.
SYSTem:COMMUnicate:SOCKeT:ADDReSS?	All	Query the IPv4 address set by the active network configuration.
SYSTem:COMMUnicate:SOCKeT:APPLY	All	Apply and save the pending network configuration.
SYSTem:COMMUnicate:SOCKeT:HostNAME?	All	Query the device hostname.
SYSTem:COMMUnicate:SOCKeT:MAC?	All	Query the device MAC address.
SYSTem:COMMUnicate:SOCKeT:NetStAt?	All	Report the current network status.
SYSTem:COMMUnicate:SOCKeT:SubnetMASK <mask>	All	Set the IPv4 subnet for the pending network configuration.
SYSTem:COMMUnicate:SOCKeT:SubnetMASK?	All	Query the IPv4 subnet set by the active network configuration.

- 4(1, 2) The behavior of this command changes depending on the model.
- 5 Some features of this command are only available to *Trains and Frames* models.

Global Commands

*IDN?

Description: Identify the device.

Response: Responds with a series of comma-separated values in SCPI fashion, representing manufacturer, model, serial number, and firmware version. This response looks similar to: HTI, T660-1, 00000, 28E662-1-1.0.

*RST

Description: Reboot the device.

Performs a soft reboot (power-on reset); it is not possible to perform a hard reset without removing the unit from power and subsequently reconnecting power. This is the only command that does not respond with *anything*, including an okay status. Upon completion of this command, either the unit is rebooted or an error is returned.

Response: This command should never provide a response.

Errors:

- ?11 Software error - If for some reason the software was unable to perform a power-on reset, this command returns with a software error. Please report any such errors encountered, as this should never occur.

Notes:

- Rebooting a unit where this command is not at the beginning of a command line does execute all prior commands on the line, however command responses of the same line may not be sent back via the issuing session before the reboot occurs. For this reason it is recommended that this command be the only command on its line.
- Behavior when multiple command sessions are issuing commands simultaneously is undefined. However, the device should still reboot.

STArT

Description: Allow triggers and resume pulse generation functionality.

Allows triggering of the device, to reverse the effects of the STOp command.

Response: OK or an error.

Notes:

- Triggers are allowed at power-up, so running this command is unnecessary unless the STOp command was previously run; this command will have no effect otherwise.

STOp

Description: Stop outputting pulses and responding to triggers.

Upon execution, this command disables all future triggers and immediately stops any running shot (by forcing an end-of-delay), causing all channel outputs to become logical low.

Response: OK or an error.

Notes:

- The effects of this command can only be reversed by using the STArT command. This means that this operation cannot be reversed in a T560-style session.
- This command terminates any possibly-running shot by issuing an end-of-delay (see: fEOD).

INSTall

Description: Apply pending settings immediately.

Queues the pending timing and channel configuration (consisting of channel configuration, timing settings, and pulse trains configuration) to be applied to the active configuration, then forces an immediate end-of-delay, terminating any possible running shot and applying the queued configuration. Issuing this command is functionally equivalent to issuing a QUEue command followed by a fEOD command.

Response: OK or an error.

Notes:

- This command **is** equivalent in function to the T560's INStall command.
- This command modifies the queued configuration.
- Running this command terminates any possibly-running shot.

QUEue

Description: Queue settings to be installed at next EOD.

Queues the pending timing and channel configuration (consisting of channel configuration, timing settings, and pulse trains configuration) to be applied to the active configuration at the next end-of-delay.

Response: OK or an error.

Notes:

- This command **is** equivalent in function to the T560's QUEue command.
- This command modifies the queued configuration, and requires an EOD (manual or at the end of a shot) to take effect.

fEOD

Description: Force an end-of-delay to occur.

Forces an end-of-delay, terminating any possibly-running shot. This applies any queued configuration settings, even if a shot was not in progress.

Response: OK or an error.

Notes:

- When run while the frames engine is active, this command advances frames engine operation as though a shot completed normally, even if a shot was not in progress.

Session Commands

ZMode <mode>

Description: Set the current command session type.

Gets or sets the current session's command parsing mode. Upon successfully changing the command parsing mode, terminates execution of the current command line immediately. This means that this command, as a command, should not have any commands following it on a single command line!

Parameters:

- <mode> - the mode to set the current command session to, one of {P500, T560}. Note that only the first two characters of each value is required.

Response: OK, or an ?07: Illegal operation error if the session type was unable to be changed.

Notes:

- Not all command sessions can switch between parsing modes. In these cases, this command will produce a ?07: Illegal operation error.
- If the command session mode was not changed, this command does not terminate the current line after execution and yields an OK response.

ZMode?

Description: Query the current command session type.

Response: Either P500 or T560 to indicate which command mode the current session is currently operating as.

Notes:

- In T560-style sessions, a trailing ? is unnecessary. However, it is recommended to always use a trailing ? character because this syntax is compatible with both P500- and T560-style sessions.

SESSion:WAIT <us>

Description: Wait a specified number of microseconds.

Parameters:

- <us> - The number of microseconds to wait, as an unsigned 32-bit integer.

Response: OK, after the wait has completed.

Notes:

- While a wait command is executing, other sessions are allowed to execute commands.

SESSion:VERBosity[:ERRor] <level>

Description: Set the session error verbosity level.

Sets the error verbosity level of the current session, leaving other sessions unchanged. This command takes effect immediately and affects all future commands, even those in the same command line.

When an error occurs as a result of any command in the current session, diagnostic information about the error is sent as a response instead of the command's normal response. This information may include:

1. At least one question mark character (?)
2. The error code, as a two digit decimal integer
3. The error name
4. A brief description of the error

Error verbosity is represented as an integer, lower values representing less information. A table of verbosity levels is provided below:

Level	Information	Example
0	The text ??	??
1	Error code	?32
2	Error code and name	?32: Unexpected argument
3	Error code, name, and brief description	?32: Unexpected argument: Too many arguments provided for the specified command.

Parameters:

- <level> - The verbosity level to set, as an integer. Only values 0 through 3 (inclusive) are valid. Values higher than 3 are interpreted as the highest available level.

Response: OK or an error.

Notes:

- Session verbosity is a per-session configuration; multiple sessions may have different values, and configuration does not persist across reboots.
- When session mode is changed via the ZMode <mode> command, error verbosity is reset to its default value for the new session type (0 for T560-style, and 3 for P500-style), so this command may need to be re-run after session type changes.
- P500-style sessions default to an error verbosity of 3, while T560-style sessions default to a verbosity of 0.

SESSion:VERBosity[:ERRor]?

Description: Query the session error verbosity level.

Response: Responds with the current session's error verbosity setting, as a 16-bit unsigned integer (0 through 3, inclusive). See the SESSion:VERBosity[:ERRor] <level> command for definitions. Values greater than 3 may be returned if they were set with the SESSion:VERBosity[:ERRor] <level> and represent the highest available verbosity level.

System Status Commands

SYSTem:ERRor:FLAGs <n>

Description: Clear global error flags.

Clears all error flags reported by the `SYSTEM:ERROR:FLAGS?` command. Persistent errors, such as the VCXO lock flag, while cleared temporarily, will be re-asserted after an undefined amount of time (less than one second) if they persist.

Parameters:

- `<n>` - May be any 32-bit integral number. This number is ignored.

Response: OK or an error.

`SYSTEM:ERROR:FLAGS?`

Description: Query global error flags.

Response: Responds with the global errors flags as a bitmask of a 16-bit unsigned integer.

Errors, their bit positions (0 to 15), and their abbreviated names as reported by this command are:

Bit	Name	Description
0	XTRIM	Oscillator trim failure.
1	RECALL	Failed to recall saved configuration. May occur at system power-on or by execution of the MEMORY:LOAD[:NORMaL] command.
2	CALIB	Invalid or lost calibration. May occur at system power-on.
3	LOGIC	Internal logic error. Asserted if the software ever encounters an internal error resulting in a potentially unsafe or undefined system state.
4	VCXO	VCXO lock error (self-clearing). Indicates a failure to lock to an external 10 MHz clock.
5	TUNE	Oscillator startup tune error. Indicates that the shot oscillator failed to calibrate at device power-on.
6	DPLL	Oscillator runtime tune error. Indicates that the shot oscillator was unable to compensate for real-time frequency variation. This is typically caused by an out-of-spec 10 MHz reference frequency or extreme ambient temperatures.
7	POWER	Power supply monitoring error. Indicates that the background built-in-self-test (BIST) function measured one or more on-board power supplies as beyond allowed voltage tolerance. This can indicate a hardware failure, or be caused by extreme ambient temperatures.
8	OLOAD	Channel output overload. Asserted while the front-panel channel output connectors are connected to a load that is drawing too much current, forcing them into a protection state.
9	TRATE	Trigger rate warning. Asserted for a short time while any triggers are received that would otherwise cause a shot to occur, but a shot was already in progress.
10	TEMP0	Active temperature compensation of the internal oscillator failed due to the temperature being measured as outside of the correctable range.
11	TEMP1	Active temperature compensation of insertion delay failed due to the temperature being measured as outside of the correctable range.

SYSTEM:BIST:POWER?

Description: Query the comprehensive device power supply status.

Reports the status of internal power supplies checked by the background built-in self-test functionality.

ly measurement statuses, expressed as "<name>", <nominal>, <measured>, <status> for each power supply, where:

ange, 1 for below minimum acceptable limit, or 2 for above maximum acceptable limit

e:

, 0, "+1.8V", 1.797, 1.802, 0, "+3.0V", 3.000, 3.019, 0, "+4Q", 3.998, 4.020, 0, "+5V", 5.043, 4.970, 0, "+10Q", 10.07

Notes:

- Not all revisions and systems share the same power supplies, and the order and quantity of responses within this command is subject to change with firmware version. As such, it is recommended to use the SYSTem:BIST:STATus? command for checking the status of background built-in self-test functions rather than parsing the data returned by this command.

SYSTem:BIST:STATus?

Description: Query the background BIST status.

Response: Responds with PASS or FAIL, indicating if the background built-in self-test functionality currently detects any power supply or processor hardware issues.

SYSTem:TEMPerature[:ALL]?

Description: Query the CPU and PCB temperatures, in degrees Celsius.

Response: Responds with the chip temperature, followed by board temperature in degrees Celsius. This response is formatted as XX.XXX, XX.XXX, with optional negative signs for each number and additional digits where necessary.

SYSTem:TEMPerature:BOARD?

Description: Query the PCB temperature, in degrees Celsius.

Response: Responds with the board temperature in degrees Celsius. This response is formatted as XX.XXX, with an optional negative sign and additional digits when necessary.

SYSTem:TEMPerature:CHIP?

Description: Query the CPU temperature, in degrees Celsius.

Response: Responds with the onboard chip temperature in degrees Celsius. This response is formatted as XX.XXX, with an optional negative sign and additional digits when necessary.

SYSTem:MICROSeconds <n>

Description: Clear the microsecond counter.

Resets the microsecond counter to 0 upon execution.

Parameters:

- <n> - May be any 32-bit integral number. This number is ignored.

Response: OK or an error.

SYSTem:MICROSeconds?

Description: Query the microsecond counter.

Response: Responds with the number of microseconds that have elapsed since the microsecond counter was last cleared or the device was last powered on or rebooted, formatted as an unsigned 32-bit integer.

FEATure:FRAMe?

Description: Query whether the frames feature is available on the device.

Response: Responds 1 if the trains and frames feature is accessible by this device, or 0 otherwise.

FEATure:STATus:FRAMe?

Description: Query whether the frames feature is available on the device.

Response: Responds 1 if the trains and frames feature is accessible by this device, or 0 otherwise.

Channel Commands

CHANnel:COMmit

Description: Apply pending timing settings immediately.

Queues the pending channel configuration (consisting of channel configuration, but excluding timing settings and pulse trains configuration) to be applied to the active configuration, then forces an immediate end-of-delay, terminating any possible running shot and applying the queued configuration. Issuing this command is functionally equivalent to issuing a CHANnel:SCHedule command followed by a fEOD command.

Response: OK or an error.

Notes:

- This command is **not** equivalent in function to the T560's INStall command. See the INSTall command.
- This command modifies the queued configuration.
- Running this command terminates any possibly-running shot.

CHANnel:SCHedule

Description: Queue all pending timing settings.

Queues the pending channel configuration (consisting of channel configuration, but excluding timing settings and pulse trains configuration) to be applied to the active configuration at the next end-of-delay.

Response: OK or an error.

Notes:

- This command is **not** equivalent in function to the T560's QUEue command. See the QUEue command.
- This command modifies the queued configuration, and requires an EOD (manual or at the end of a shot) to take effect.

CHANnel:<stage>:MODE <channel>, <enable>

Description: Enable or disable a channel's output.

Enables or disables the output of a channel by modifying the channel's pending configuration. When a channel is set to be output disabled, it outputs a low-impedance logical low signal in accordance with its termination and polarity settings. This command does not affect a channel's termination nor its polarity.

Immediately after changing the pending configuration, a channel configuration install or queue may be performed, depending on the value of <stage>:

- If <stage> is ACTive, then a CHANNEL:COMmit will be performed. The pending, queued, and active configurations are modified. This will immediately force an end-of-delay.
- If <stage> is NEXT, then a CHANNEL:SCHedule will be performed. Both the pending and queued configurations are modified. The active configuration will be updated to the new queued configuration at the next end-of-delay.
- If <stage> is QUEue, then only the pending configuration will be modified. An install or queue operation will need to be run for any changes to take effect.

Parameters:

- <stage> - May be ACTive, NEXT, or QUEue to target the active, queued, or pending configurations respectively. NEXT is only accessible with the *Trains and Frames* feature.
- <channel> - The channel to modify settings of. This may be one of A, B, C, D.
- <enable> - Either ON if the output is to be enabled, or OFF if it is to be disabled.

Response: OK or an error.

CHANNEL:<stage>:MODE? <channel>

Description: Query the output enable state of a channel.

Parameters:

- <stage> - May be ACTive, NEXT, or QUEue to target the active, queued, or pending configurations respectively. NEXT is only accessible with the *Trains and Frames* feature.
- <channel> - The channel to query settings of. This may be one of A, B, C, D.

Response: Reads the specified configuration (active, queued, or pending), then responds with either ON or OFF, corresponding to output enabled and disabled, respectively.

CHANNEL:<stage>:POLarity <channel>, <polarity>

Description: Configure a channel's output polarity.

Sets the output state of a channel to be either active-high (positive) or active-low (negative) by modifying the channel's pending configuration.

Immediately after changing the pending configuration, a channel configuration install or queue may be performed, depending on the value of <stage>:

- If <stage> is ACTive, then a CHANNEL:COMmit will be performed. The pending, queued, and active configurations are modified. This will immediately force an end-of-delay.
- If <stage> is NEXT, then a CHANNEL:SCHedule will be performed. Both the pending and queued configurations are modified. The active configuration will be updated to the new queued configuration at the next end-of-delay.
- If <stage> is QUEue, then only the pending configuration will be modified. An install or queue operation will need to be run for any changes to take effect.

Parameters:

- <stage> - May be ACTIVE, NEXT, or QUEUE to target the active, queued, or pending configurations respectively. NEXT is only accessible with the *Trains and Frames* feature.
- <channel> - The channel to modify settings of. This may be one of A, B, C, D.
- <polarity> - Either POSITIVE if the output is to be set active-high, or NEGATIVE for active-low.

Response: OK or an error.

CHANNEL:<stage>:POLARITY? <channel>

Description: Query a channel's output polarity.

Parameters:

- <stage> - May be ACTIVE, NEXT, or QUEUE to target the active, queued, or pending configurations respectively. NEXT is only accessible with the *Trains and Frames* feature.
- <channel> - The channel to query settings of. This may be one of A, B, C, D.

Response: Reads the specified configuration (active, queued, or pending), then responds with either POSITIVE if the channel is set to be active-high, or NEGATIVE if active-low.

CHANNEL:<stage>:TERMINATION <channel>, <termination>

Description: Configure a channel's output termination.

Sets the termination of a channel's connector to be either low-impedance or 50 Ohm source terminated by modifying the channel's pending configuration.

Immediately after changing the pending configuration, a channel configuration install or queue may be performed, depending on the value of <stage>:

- If <stage> is ACTIVE, then a CHANNEL:COMMIT will be performed. The pending, queued, and active configurations are modified. This will immediately force an end-of-delay.
- If <stage> is NEXT, then a CHANNEL:SCHEDULE will be performed. Both the pending and queued configurations are modified. The active configuration will be updated to the new queued configuration at the next end-of-delay.
- If <stage> is QUEUE, then only the pending configuration will be modified. A install or queue operation will need to be run for any changes to take effect.

Parameters:

- <stage> - May be ACTIVE, NEXT, or QUEUE to target the active, queued, or pending configurations respectively. NEXT is only accessible with the *Trains and Frames* feature.

Parameters:

- <channel> - The channel to modify settings of. This may be one of A, B, C, D.
- <termination> - The output termination to apply to the channels' connectors. Either LOWZ for low-impedance or 50OHM for 50 Ohm source termination.

Response: OK or an error.

Notes:

- This command terminates any possibly-running shot by issuing an end-of-delay (see: FEOD).

CHANNEL:<stage>:TERMINATION? <channel>

Description: Query a channel's output termination.

Parameters:

- <stage> - May be ACTIVE, NEXT, or QUEUE to target the active, queued, or pending configurations respectively. NEXT is only accessible with the *Trains and Frames* feature.
- <channel> - The channel to query settings of. This may be one of A, B, C, D.

Response: Reads the specified configuration (active, queued, or pending), then responds with NONE if the channel output is configured to be low-impedance, or 500HM if source-terminated with 50 Ohms.

CHANnel:TimingMODE <channel>, <timing_mode>

Description: Set a channel's timing mode.

Sets a channel's timing mode to be either delay-width or rise-fall. In delay-width mode, the affected channel(s) falling edge cannot be referenced to a different edge. Thus, when applying delay-width mode to a channel, the falling edge will be changed to reference the rising edge of the same channel, while the rising edge will keep its reference edge. Whenever a channel's timing mode settings are modified, all delays, relative to trigger assertion, remain the same as they were prior to the command.

See [Channel Timing Mode](#) for additional details.

Parameters:

- <channel> - The channel to modify settings of. This may be one of A, B, C, D.
- <timing_mode> - Which timing mode the channel(s) will operate using, either DelayWidth for delay-width or RiseFall for rise-fall.

Response: OK or an error.

CHANnel:TimingMODE? <channel>

Description: Query a channel's timing mode.

Parameters:

- <channel> - The channel to query settings of. This may be one of A, B, C, D.

Response: Responds with which timing mode the channel's edge timings are currently operating under, either DelayWidth for delay-width or RiseFall for rise-fall.

CHANnel:ON <channel>

Description: Enable a channel's output.

Equivalent to CHANnel[:ACTIVE]:MODE <channel>, <enable> with a <enable> parameter value of ON.

Parameters:

- <channel> - The channel to modify settings of. This may be one of A, B, C, D.

Response: OK or an error.

Notes:

- This command terminates any possibly-running shot by issuing an end-of-delay (see: FEOD).

CHANnel:ON? <channel>

Description: Query the output enable state of a channel.

Equivalent to CHANnel[:ACTIVE]:MODE? <channel>.

Parameters:

- <channel> - The channel to query settings of. This may be one of A, B, C, D.

Response: Responds with ON or OFF, ON corresponding to the channel's output being enabled, and OFF with the channel's output disabled.

CHANnel:OFF <channel>

Description: Disable a channel's output.

Equivalent to CHANnel[:ACTive]:MODE <channel>, <enable> with a <enable> parameter value of OFF.

Parameters:

- <channel> - The channel to modify settings of. This may be one of A, B, C, D.

Response: OK or an error.

Notes:

- This command terminates any possibly-running shot by issuing an end-of-delay (see: fEOD).

CHANnel:OFF? <channel>

Description: Query the output enable state of a channel.

Equivalent to CHANnel[:ACTive]:MODE? <channel>.

Parameters:

- <channel> - The channel to query settings of. This may be one of A, B, C, D.

Response: Responds with ON or OFF, ON corresponding to the channel's output being enabled, and OFF with the channel's output disabled.

CHANnel:POSitive <channel>

Description: Set a channel's output polarity to be active-high (positive)

Equivalent to CHANnel[:ACTive]:POLarity <channel>, <polarity> with a <polarity> parameter value of POSitive.

Parameters:

- <channel> - The channel to modify settings of. This may be one of A, B, C, D.

Response: OK or an error.

Notes:

- This command terminates any possibly-running shot by issuing an end-of-delay (see: fEOD).

CHANnel:POSitive? <channel>

Description: Query the output polarity of a channel.

Equivalent to CHANnel[:ACTive]:POLarity? <channel>.

Parameters:

- <channel> - The channel to query settings of. This may be one of A, B, C, D.

Response: Responds with POSitive if the channel is currently active-high, or NEGative if active-low.

CHANnel:NEGative <channel>

Description: Set a channel's output polarity to be active-low (negative)

Equivalent to CHANnel[:ACTive]:POLarity <channel>, <polarity> with a <polarity> parameter value of NEGative.

Parameters:

- <channel> - The channel to modify settings of. This may be one of A, B, C, D.

Response: OK or an error.

Notes:

- This command terminates any possibly-running shot by issuing an end-of-delay (see: fEOD).

CHANnel:NEGative? <channel>

Description: Query the output polarity of a channel.

Equivalent to CHANnel[:ACTive]:POLarity? <channel>.

Parameters:

- <channel> - The channel to query settings of. This may be one of A, B, C, D.

Response: Responds with POSitive if the channel is currently active-high, or NEGative if active-low.

CHANnel:DelayWidth <channel>

Description: Set a channel's timing mode to be delay-width.

Equivalent to CHANnel:TimingMODE <channel>, <timing_mode> with a <timing_mode> parameter value of DelayWidth.

Parameters:

- <channel> - The channel to modify settings of. This may be one of A, B, C, D.

Response: OK or an error.

CHANnel:DelayWidth? <channel>

Description: Query the timing mode of a channel.

Equivalent to CHANnel:TimingMODE? <channel>.

Parameters:

- <channel> - The channel to query settings of. This may be one of A, B, C, D.

Response: Responds with which timing mode the channel's edge timings are currently operating under, either DelayWidth for delay-width or RiseFall for rise-fall.

CHANnel:RiseFall <channel>

Description: Set a channel's timing mode to be rise-fall.

Equivalent to CHANnel:TimingMODE <channel>, <timing_mode> with a <timing_mode> parameter value of RiseFall.

Parameters:

- <channel> - The channel to modify settings of. This may be one of A, B, C, D.

Response: OK or an error.

CHANnel:RiseFall? <channel>

Description: Query the timing mode of a channel.

Equivalent to CHANnel:TimingMODE? <channel>.

Parameters:

- <channel> - The channel to query settings of. This may be one of A, B, C, D.

Response: Responds with which timing mode the channel's edge timings are currently operating under, either DelayWidth for delay-width or RiseFall for rise-fall.

CHANnel:500HM <channel>

Description: Enable a channel's output termination.

Sets the termination of a channel's connector to be 50 Ohm source terminated.

Equivalent to CHANnel[:ACTive]:TERmination <channel>, <termination> with a <termination> parameter value of 500HM.

Parameters:

- <channel> - The channel to modify settings of. This may be one of A, B, C, D.

Response: OK or an error.

Notes:

- This command terminates any possibly-running shot by issuing an end-of-delay (see: fEOD).

CHANnel:500HM? <channel>

Description: Query a channel's output termination.

Equivalent to CHANnel[:ACTive]:TERMination? <channel>.

Parameters:

- <channel> - The channel to query settings of. This may be one of A, B, C, D.

Response: Responds with NONE if the channel output is low-impedance or 500HM if source-terminated with 50 Ohms.

CHANnel:LOWZ <channel>

Description: Disable a channel's output termination.

Sets the termination of a channel's connector to be low-impedance.

Equivalent to CHANnel[:ACTive]:TERMination <channel>, <termination> with a <termination> parameter value of LOWZ.

Parameters:

- <channel> - The channel to modify settings of. This may be one of A, B, C, D.

Response: OK or an error.

Notes:

- This command terminates any possibly-running shot by issuing an end-of-delay (see: fEOD).

CHANnel:LOWZ? <channel>

Description: Query a channel's output termination.

Equivalent to CHANnel[:ACTive]:TERMination? <channel>.

Parameters:

- <channel> - The channel to query settings of. This may be one of A, B, C, D.

Response: Responds with NONE if the channel output is low-impedance or 500HM if source-terminated with 50 Ohms.

Timing Commands

TIME:COMmit

Description: Apply pending timing settings immediately.

Queues the pending timing configuration (consisting of timing settings and pulse trains configuration, but excluding channel configuration) to be applied to the active configuration, then forces an immediate end-of-delay, terminating any possible running shot and applying the queued configuration. Issuing this command is functionally equivalent to issuing a TIME:SCHedule command followed by a fEOD command.

Response: OK or an error.

Notes:

- This command is **not** equivalent in function to the T560's `INStall` command. See the `INSTall` command.
- This command modifies the queued configuration.
- Running this command terminates any possibly-running shot.

TIME:SCHedule

Description: Queue all pending timing settings.

Queues the pending timing configuration (consisting of timing settings and pulse trains configuration, but excluding channel configuration) to be applied to the active configuration at the next end-of-delay.

Response: OK or an error.

Notes:

- This command is **not** equivalent in function to the T560's `QUEue` command. See the `QUEue` command.
- This command modifies the queued configuration, and requires an EOD (manual or at the end of a shot) to take effect.

TIME:DElay<edge> <de lay>

Description: Set and install timing edge's delay.

Overwrites the pending configuration edge delay, then commits the timing configuration. This is equivalent to running the `TIME:QUEue<edge> <de lay>`, then `TIME:COMmit` commands.

Parameters:

- `<edge>` - The index of the channel edge to configure; values 1 through 8 (inclusive) are accepted. See [Timing Edges](#) for definitions.
- `<de lay>` - The delay, relative to the specified edge's reference edge, as a time. Negative numbers are allowed, depending on the configured reference edge. See [Time Formatting](#) for formatting requirements.

Response: OK or an error.

Notes:

- This command is affected by the edge's reference edge configuration; see the [Edge Relativity](#) command for details.
- Edge relativity is calculated based on the pending, not the active timing delays, just like the `TIME:QUEue<edge> <de lay>` command.

TIME:DElay<edge>?

Description: Query a timing edge's active delay.

Parameters:

- `<edge>` - The index of the channel edge to query information from. Values 1 through 8 (inclusive) are accepted. See [Timing Edges](#) for definitions.

Response: Responds with the current active delay of the specified edge, as a time (see [Time Formatting](#) for formatting). It is possible for this response to include a negative number if the edge's reference edge configuration allows it.

Notes:

- This command is affected by the edge's reference edge configuration; see the [Edge Relativity](#) command for details.

TIME:INSDel <mode>

Description: Set the insertion delay mode.

Sets the insertion delay mode. The T660 currently only supports fast mode, however supports this command for P500 script compatibility. Because of this, this command has no effect.

Parameters:

- <mode> - FAST or NORMaL. Currently, only FAST is allowed. Using a value of NORMaL will produce a ?06: Feature required error.

Response: OK or an error.

TIME:INSDeL?

Description: Query the insertion delay mode.

Response: Either FAST or NORMAL, corresponding to the currently configured insertion delay mode. Since the T660 only supports fast mode, it will always respond FAST.

TIME:NEXT<edge>

Description: Set and queue a timing edge delay change.

Overwrites the pending configuration edge delay, then issues a queue operation. This is equivalent to running the TIME:QUEue<edge> <deLay>, then TIME:SCHedule commands.

Parameters:

- <edge> - The index of the channel edge to configure; values 1 through 8 (inclusive) are accepted. See [Timing Edges](#) for definitions.
- <deLay> - The delay, relative to the specified edge's reference edge, as a time. Negative numbers are allowed, depending on the configured reference edge. See [Time Formatting](#) for formatting requirements.

Response: OK or an error.

Notes:

- This command is affected by the edge's reference edge configuration; see the [Edge Relativity](#) command for details.
- Edge relativity is calculated based on the pending, not the queued timing delays, just like the TIME:QUEue<edge> <deLay> command.

TIME:NEXT<edge>?

Description: Query a timing edge's queued delay.

Parameters:

- <edge> - The index of the channel edge to query information from. Values 1 through 8 (inclusive) are accepted. See [Timing Edges](#) for definitions.

Response: Responds with the current pending delay of the specified edge, as a time (see [Time Formatting](#) for formatting). It is possible for this response to include a negative number if the edge's reference edge configuration allows it.

Notes:

- This command is affected by the edge's reference edge configuration; see the [Edge Relativity](#) command for details.

TIME:QUEue<edge> <deLay>

Description: Set a timing edge's pending delay.

Overwrites the pending configuration edge delay. Edge relativity is calculated relative to the current pending configuration.

A channel's rising edge must occur after its falling edge in absolute time. If this condition would be invalidated by the execution of this command, this command will produce a ?08: Settings conflict error.

Parameters:

- <edge> - The index of the channel edge to configure; values 1 through 8 (inclusive) are accepted. See [Timing Edges](#) for definitions.
- <delay> - The delay, relative to the specified edge's reference edge, as a time. Negative numbers are allowed, depending on the configured reference edge. See [Time Formatting](#) for formatting requirements.

Response: OK or an error.

Notes:

- This command is affected by the edge's reference edge configuration; see the [Edge Relativity](#) command for details.

TIME:QUEue<edge>?

Description: Query a timing edge's pending delay.

Parameters:

- <edge> - The index of the channel edge to query information from. Values 1 through 8 (inclusive) are accepted. See [Timing Edges](#) for definitions.

Response: Responds with the current pending delay of the specified edge, as a time (see [Time Formatting](#) for formatting). It is possible for this response to include a negative number if the edge's reference edge configuration allows it.

Notes:

- This command is affected by the edge's reference edge configuration; see the [Edge Relativity](#) command for details.

TIME:RELTo<edge> <reference_edge>

Description: Set a timing edge's reference edge.

Configures all commands and queries for the specified channel edge to be interpreted and reported as relative to another timing edge. When a timing command is run, it references the current absolute delay for an edge to calculate delay relative to a time argument or response.

See [Edge Relativity](#) for additional details on how this configuration works.

Parameters:

- <edge> - The index of the channel edge to configure; values 1 through 8 (inclusive) are accepted. See [Timing Edges](#) for definitions.
- <reference_edge> - The timing edge that delay will be measured relative to, a number 0 through 8 (inclusive). See [Timing Edges](#) for definitions. This value may not be the same as <edge>.

Response: OK or an error.

Notes:

- While a channel is in delay-width mode, the falling edge for a channel is immutable and fixed to the rising edge of a channel. In this case, this command has no effect and does not modify configuration, but will still produce an OK response. See the CHANnel:TimingMOde <channel>, <timing_mode> command for details.

TIME:RELTo<edge>?

Description: Query a timing edge's reference edge.

Parameters:

- <edge> - The index of the channel edge to query information from. Values 1 through 8 (inclusive) are accepted.

Response: A number 0 through 8 (inclusive), corresponding to the timing edge that commands and queries for the specified channel edge are relative to. See [Timing Edges](#) for definitions.

TIME:UNDO

Description: Revert changes to the pending configuration.

Loads the queued timing configuration (consisting of timing settings and pulse trains configuration, but excluding channel configuration) into the pending configuration.

Response: OK or an error.

Notes:

- This command modifies the pending configuration, and requires a queue or install to take effect.

Trigger Commands

TRIGger:EXECute

Description: Fire a remote trigger.

Executes a remote trigger only if the trigger source is set to remote. If it is not, this command produces an error (specifically, a ?07: Illegal operation error).

Response: OK or an error.

Notes:

- This command can only be executed with certain global configurations, and has normal behavior that includes producing a ?07: Illegal operation error if this requirement is not met.
- Remote triggers are affected by predivider and burst configuration just like other trigger sources.

TRIGger[:EXternal]:PREDiv <divisor>

Description: Query the trigger pre-divider divisor value.

Sets the number of triggers that should be received before the next trigger is considered valid. A value of 0 is functionally equivalent to a value of 1, and indicates that the predivider is disabled. This configuration applies to all trigger sources, even remote triggers.

Parameters:

- <divisor> - The trigger divisor value to set, as a 32-bit unsigned integer. Hexadecimal notations are allowed, and the full range of the integer is valid. 0 implies a divisor value of 1.

Response: OK or an error.

TRIGger[:EXternal]:PREDiv?

Description: Set the trigger pre-divider divisor value.

Reports the number of triggers that should be received before the next trigger is considered valid. A value of 0 is functionally equivalent to a value of 1, and indicates that the predivider is disabled.

Response: Responds with the configured predivider divisor, as a 32-bit unsigned integer.

TRIGger:FREQuency[:SYNth] <frequency>

Description: Set the frequency of the on-board trigger synthesizer.

The maximum synthesizable frequency is 16 MHz; values higher than this will produce an error.

Parameters:

- <frequency> - The frequency to set the trigger synthesizer to, in Hertz by default, as a floating-point number. Available suffixes are Mhz, Khz, and Hz. Values higher than 16 MHz or lower than 0 Hz (negative values) are not allowed.

Response: OK or an error.

Notes:

- This command uses different suffixes than the P500 does (the P500 uses MAHZ, KHZ, HZ, MHZ). This means that MHz and MAHZ will be interpreted differently on a T660 than a P500.
- Although unlikely in practice, frequency transitions are not guaranteed to be glitchless; it is possible for the trigger rate to intermediately be higher or lower than both the previous and new frequency values temporarily.
- For extremely low frequencies, it is recommended to utilize the trigger predivider in combination with the internal trigger source to reduce potentially significant jitter (see: TRIGger[:EXTeRnAL]:PREDiv <divisor>).

TRIGger:FREQuency[:SYNth]?

Description: Query the frequency of the on-board trigger synthesizer.

Reports the configured trigger synthesizer frequency. This value is not measured, but instead reported from configuration.

Response: Responds with the configured trigger synthesizer frequency, formatted as a frequency (+0000000000.000000, in Hertz).

TRIGger:FREQuency:HITs?

Description: Query the hit frequency counter.

Response: Responds with the measured hit frequency, formatted as a frequency (+0000000000.000000, in Hertz).

Notes:

- A running average over 1 second is performed to determine the frequency of events. However, periods with no activity (in roughly 8ms granularity) are discarded from this calculation so that a good estimate of frequency can be requested prior to 1 second elapsing. This can cause confusing results in the presence of intermittent triggers, so it is recommended to only use these commands with trigger sources faster than 1 kHz.

TRIGger:FREQuency:TRIGger?

Description: Query the trigger frequency counter.

Response: Responds with the measured trigger frequency, before the predivider, formatted as a frequency (+0000000000.000000, in Hertz).

Notes:

- A running average over 1 second is performed to determine the frequency of events. However, periods with no activity (in roughly 8ms granularity) are discarded from this calculation so that a good estimate of frequency can be requested prior to 1 second elapsing. This can cause confusing results in the presence of intermittent triggers, so it is recommended to only use these commands with trigger sources faster than 1 kHz.

TRIGger:PERiod[:SYNth]?

Description: Query the period of the on-board trigger synthesizer.

Response: Responds with the configured trigger synthesizer frequency, formatted as a time (+000.000000000000, in seconds). This value can never be negative.

TRIGger:PERiod:HITs?

Description: Query the hit period counter.

Response: Responds with the measured trigger hit period, formatted as a time (+000.000000000000, in seconds). This value can never be negative.

Notes:

- A running average over 1 second is performed to determine the frequency of events. However, periods with no activity (in roughly 8ms granularity) are discarded from this calculation so that a good estimate of frequency can be requested prior to 1 second elapsing. This can cause confusing results in the presence of intermittent triggers, so it is recommended to only use these commands with trigger sources faster than 1 kHz.

TRIGger:PERiod:TRIGger?

Description: Query the trigger period counter.

Response: Responds with the measured trigger period, before the predivider, formatted as a time (+000.000000000000, in seconds). This value can never be negative.

Notes:

- A running average over 1 second is performed to determine the frequency of events. However, periods with no activity (in roughly 8ms granularity) are discarded from this calculation so that a good estimate of frequency can be requested prior to 1 second elapsing. This can cause confusing results in the presence of intermittent triggers, so it is recommended to only use these commands with trigger sources faster than 1 kHz.

TRIGger:SOURce <source>

Description: Set the trigger source.

Changes the trigger source configuration to source triggers only from the specified source. Even if the current trigger source is already identical to the one requested, execution of this command will temporarily disable triggers, so triggers may be missed if repeatedly issued.

Trigger sources and their abbreviations are as follows:

Abbreviation	Name
OFF	Triggering disabled
EXT	External trigger (front panel)
INT	Internal trigger (80 MHz)
SYN	Synthesized input trigger (DDS)
REM	Remote trigger (command only)

Parameters:

- <source> - Where to source triggers from. May be one of {OFF, INT, SYN, REM, EXT} representing the configured trigger source.

Response: OK or an error.

TRIGger:SOURce?

Description: Query the current trigger source.

Response: One of {OFF, INT, SYN, REM, EXT} representing the configured trigger source. See the TRIGger:SOURce <source> command for trigger source descriptions.

TRIGger:SHOTs <n>

Description: Clear the shot counter.

Parameters:

- <n> - May be any integral number. This number is ignored.

Response: OK or an error.

TRIGger:SHOTs?

Description: Query the shot counter value.

Response: Responds with the number of shots that have elapsed since TRIGger:SHOTs <n> was last called or the device was last powered on or rebooted, formatted as an unsigned 32-bit integer.

TRIGger:INPut:POLarity <polarity>

Description: Set the external trigger polarity.

Parameters:

- <polarity> - The polarity to configure, POSitive for positive (logical high) or NEGative for negative (inverted; logical low).

Response: OK or an error.

TRIGger:INPut:POLarity?

Description: Query the external trigger polarity.

Reports the last-configured external trigger polarity, even if the current trigger source is not external.

Response: Responds with the trigger polarity, POS for positive (logical high) or NEG for negative (inverted; logical low).

TRIGger:INPut:TERMination <termination>

Description: Enable or disable trigger connector end-termination.

Enable or disable end termination of the trigger connector. When run, this command immediately engages or disengages the 50 Ohm (to ground) end-termination of the external trigger connector.

Parameters:

- <termination> - The state to set termination to. Either HIZ for low-impedance or 50OHM for 50 Ohm source termination.

Response: OK or an error.

TRIGger:INPut:TERMination?

Description: Query the trigger connector end-termination state.

Response: Either 50OHM or NONE to reflect the current end termination state of the trigger connector.

TRIGger:INPut:VOLTage <voltage>

Description: Query the external trigger voltage threshold.

Configures the external trigger connector's logic-level threshold. Voltages above this threshold are considered high, and those below low. This does not account for hysteresis.

Parameters:

- <voltage> - The threshold voltage to set, in Volts, as a floating point number. No suffixes are supported. Values lower than 0.25 Volts and higher than 3.3009 Volts are not allowed and will produce an error.

Response: OK or an error.

TRIGger:INPut:VOLTage?

Description: Set the external trigger voltage threshold.

Response: Responds with the current external trigger logic level threshold, rounded to the nearest millivolt, and formatted as +0.000. Additional leading digits are added as necessary for high voltages, and the precision is fixed to 1 millivolt.

Notes:

- Unlike the P500, the precision of this response is always significant to 1 millivolt, rather than 10.

Gate Commands

GATE:MODE <mode>

Description: Set the gate operating mode.

Gate configurations, as represented by this command, are:

<mode>	Operating Mode	Connector Polarity
0	Disabled	N/A
1	Output	Positive
2		Negative
3	Input	Positive
4		Negative
5	Burst (Input)	Positive
6		Negative
7	Auxiliary (Output)	Positive
8		Negative
9	Remote (Output)	Positive
10		Negative

Note that utilizing the gate in auxiliary mode is restricted to *Trains and Frames* models only.

When the gate connector is utilized as an output, end termination is disabled, though this does not change the gate termination configuration. A 50-Ohm source termination is always present in output modes.

When the gate connector is utilized as an input, the previously-configured gate termination is re-applied.

Parameters:

- <mode> - An integer, 0 through 10, corresponding to the desired gate configuration.

Response: OK or an error.

GATE:MODE?

Description: Query the current gate operating mode.

Response: Responds with the current gate configuration as an integer 0 through 10. See the GATE:MODE <mode> command for definitions of this number.

GATE:INPut:TERMination <termination>

Description: Enable or disable gate connector end-termination.

Configures the termination of the gate connector while not an output.

Parameters:

- <termination> - The input termination to apply to the gate connector. Either HIZ for high-impedance, or 50OHM for 50 Ohm end termination.

Response: OK or an error.

Notes:

- Termination configuration applies only when the gate mode is not set to output. This means that this configuration applies even while in gate remote mode and while gate functionality is configured as off.
- This command has no effect on hardware when run while the gate connector is utilized as an output (in output or auxiliary modes). However, the configured setting is applied immediately and will take effect once the gate connector is no longer utilized as an output.

GATE:INPut:TERMination?

Description: Query the gate connector end-termination state.

Response: NONE, or 50OHM if the gate connector is currently set to be end terminated when utilized as a non-output.

GATE:EXECute

Description: Fire a single remote burst.

Simulates the gate connector pulsing high to fire a remote burst while the device is in gate remote mode.

Response: OK or an error.

Notes:

- If this command is run while the device is not in gate remote mode (see: GATE:MODE <mode>), it will still produce the response OK, but will not perform any function.

Auxiliary Output Commands

SYSTem:AUXiliary:MODE <mode>

Description: Configure the auxiliary output operating mode.

Parameters:

- <mode> - The mode to configure the auxiliary output to, either UPDate or TRIGger, referring to update and trigger modes, respectively.

Response: OK or an error.

Notes:

- Because the T660 does not have a dedicated auxiliary output connector, this command has no visible effect on hardware unless the gate connector has been configured to output the auxiliary output using the GATE:MODE <mode> command.

SYSTem:AUXiliary:MODE?

Description: Query the auxiliary output operating mode.

Response: Either UPDATE or TRIGGER, referring to auxiliary output update and trigger modes, respectively.

SYSTem:AUXiliary:UPDate:WIDTh <time>

Description: Set the auxiliary output update mode pulse width.

Sets the width of pulses generated by the auxiliary output when in update mode. The auxiliary output, when in trigger mode, is not affected by this setting.

Parameters:

- <time> - The pulse width, as a time. Available suffixes are Ps (picoseconds), Ns (nanoseconds), Ms (milliseconds), and S (seconds). Default units are seconds. Scientific notation and SCPI-format numbers are accepted. Minimum increments are in picoseconds. Values 0 up to 3ms (inclusive) are allowed. This value is rounded down to the nearest multiple of 20 nanoseconds.

Response: OK or an error.

Notes:

- Pulse widths are rounded to the nearest multiple of 20 nanoseconds when set. Use the SYSTem:AUXiliary:UPDate:WIDTh? command to read what value was applied.
- Pulse widths not equal to a multiple of 20 nanoseconds do not generate an error, but values out of the valid range do generate errors.
- Although discouraged, a pulse width of 0 is allowed and supported. In this case, pulses will never be observed.

SYSTem:AUXiliary:UPDate:WIDTh?

Description: Query the auxiliary output update mode pulse width.

Response: Responds with the current effective auxiliary pulse width (for auxiliary update mode pulses) formatted as a time. Time values are extended-precision floating values with units of seconds, which may require more than a double-precision floating point to represent full precision.

Burst Commands

BURst:CClear

Description: Clear the burst counter.

Immediately clears both the predivider and burst counters.

Response: OK or an error.

BURst:CClear?

Description: Query the current burst counter count.

Response: Responds with an integer representing the current burst counter value. This counter value will be between 0 and the limit set by BURst:TRIGger <m_triggers> - 1, inclusive.

BURst:MODE <enable>

Description: Enable or disable burst mode.

Enables or disables burst functionality depending on the value of <enable>.

Parameters:

- <enable> - Either ON if burst mode is to be enabled, or OFF if it is to be disabled.

Response: OK or an error.

BURst:MODE?

Description: Query whether burst mode is enabled.

Response: Responds ON if burst mode is currently enabled, or OFF if disabled.

BURst:PULse <n_pulses>

Description: Set the burst pulse count.

Sets the burst 'N' value, and updates the burst configuration.

Parameters:

- <n_pulses> - The number of shots that will occur in a burst, akin to the 'numerator' of the burst logic.

Response: OK or an error.

Notes:

- Execution of this command clears *both* the predivider and burst counters.
- If N = 0, M = 0, or N > M, burst functionality is disabled. Using bursts with N = M is not advised since this may produce unintuitive behavior (bursts will be enabled if burst is configured as on).
- Operating with the gate operation set to remote or gate burst mode with burst disabled (invalid N and M values satisfy this) behaves as though burst is enabled, with N = 1 and M = 1.

BURst:PULse?

Description: Query the burst pulse count.

Response: Responds with the burst 'N' value as a 32-bit unsigned integer.

BURst:TRIGger <m_triggers>

Description: Set the burst period count.

Sets the burst 'M' value, and updates the burst configuration.

Parameters:

- <m_triggers> - The number of shots that compose a burst, akin to the 'denominator' of the burst logic.

Response: OK or an error.

Notes:

- Execution of this command clears *both* the predivider and burst counters.
- If $N = 0$, $M = 0$, or $N > M$, burst functionality is disabled. Using bursts with $N = M$ is not advised since this may produce unintuitive behavior (bursts will be enabled if burst is configured as on).
- Operating with the gate operation set to remote or gate burst mode with burst disabled (invalid N and M values satisfy this) behaves as though burst is enabled, with $N = 1$ and $M = 1$.

BURst:TRIGger?

Description: Query the burst period count.

Response: Responds with the burst 'M' value as a 32-bit unsigned integer.

Clock Commands

CLOCK:MODE <mode>

Description: Set the clock connector operating mode.

Parameters:

- <mode> - OFF to disable the clock connector, OUTput to utilize the clock as an output, or INPut to utilize it as an input.

Response: OK or an error.

CLOCK:MODE?

Description: Query the clock connector operating mode.

Response: OFF if the clock connector is disabled, OUT if currently utilized as an output, and INP if currently utilized as an input.

CLOCK:EXTeRnAL <enable>

Description: Enable or disable locking to an external clock.

While enabled, the T660 will attempt to lock to an external clock signal. If unable to do so, this setting shall be unchanged, however the T660 will behave as though this setting were disabled. If an external oscillator signal is lost, the T660 will resume attempting to lock while this setting is enabled.

Parameters:

- <enable> - ON (or 1) to enable external oscillator locking functionality, or OFF (or 0) to disable it. Note that these accepted numerical aliases are strings, not numbers, so 01 is not a valid value.

Response: OK or an error.

CLOCK:EXTeRnAL?

Description: Query whether external clock locking is enabled.

Response: Responds with 1 if the clock connector is currently utilized as an input and the device is attempting to lock to an external 10 MHz clock, or 0 otherwise.

CLOCK:FREQuency?

Description: Query the perceived external clock frequency.

Response: Responds with the clock frequency observed at the clock connector, relative to the internal frequency reference, expressed as an unsigned integer with units of Hertz.

Notes:

- If the clock connector is set to lock to an external source, the response value should be fixed at *10000000*, indicating that an exact 10 MHz lock exists. A value close to but not exactly equal to 10 MHz may indicate an unstable clock, or that the external clock is a frequency outside the lockable range of the device.
- If the clock connector is set to output a 10 MHz clock, a response value other than *10000000* may indicate a problematic hardware condition.

CLOCK:OCX0?

Description: Query whether an OCXO is installed.

Response: Responds with 1 to indicate that this device has an OXCO (Oven Controlled Crystal Oscillator) installed, or 0 otherwise. The T660 will always report 0 because it does not support an OCXO add-on.

CLOCK:STATUs?

Description: Query the status of locking to an external clock.

Response: Responds INTL, if the clock connector is disabled or utilized as an output. Otherwise, if the clock connector is attempting to lock to an external 10 MHz clock (wherein the clock connector is an input), responds with one of the following: NOTPRESENT, to indicate that no clock was detected at the connector, NOTLOCKED, indicating that a signal is present but the internal PLL has not locked to the external clock, or LOCKED to indicate a successful lock to an external clock.

CLOCK:TRIM[:SET] <trim>

Description: Set the VCXO trim value.

Parameters:

- <trim> - The active trim value for the on-board oscillator, as a 16-bit unsigned integer. A value difference of 1 affects the oscillator frequency by approximately 7 ppt (parts per trillion), however this is not specified or guaranteed across units.

Response: OK or an error.

Notes:

- This parameter requires the CLOCK:TRIM:SAVE command to be loaded at power-up.
- Adjusting the VCXO trim too far from its initial value at power-up (further than the typical external reference locking range) can cause excessive output jitter. If this occurs, the user-configured trim value must be saved and the unit rebooted.
- Changes are applied immediately to hardware. If the clock is currently being used as an input, the effect of this command will not be visible until the clock mode is changed to output or disabled.

CLOCK:TRIM[:SET]?

Description: Query the effective VCXO trim value.

Reports the active VCXO trim value (not the value saved in non-volatile memory). If a user-mandated trim value has been set, this will override the factory-calibrated default trim value.

Response: The active trim value for the on-board oscillator, as a 16-bit unsigned integer. A value difference of 1 affects the oscillator frequency by approximately 7 ppt (parts per trillion), however this is not specified or guaranteed across units.

Notes:

- This command only reports the configuration status, not what the VCXO trim value is while locking to an external 10 MHz reference.

CLOCK:TRIM:DEFault

Description: Restore the factory default VCXO trim value.

Loads and applies the factory-calibrated VCXO trim value (just like the CLOCK:TRIM[:SET] <trim> command does). This command does not save the loaded trim value; use the CLOCK:TRIM:SAVE command to save this value to non-volatile memory.

Response: OK or an error.

CLOCK:TRIM:SAVE

Description: Save the user-set VCXO trim value to nonvolatile memory.

Save the user-set VCXO trim value to non-volatile memory, to be loaded at every power-up in place of the factory-calibrated trim value.

Response: OK or an error.

Trains Commands

TRAI:n:<stage>:CouNT <pulses>

Description: Set the number of train pulses per trigger.

Sets the trains pulse count of the pending configuration. A value of 0 indicates that trains will be disabled.

Immediately after changing the pending configuration, a timing configuration install or queue may be performed, depending on the value of <stage>:

- If <stage> is ACTIVE, then a TIME:COMmit will be performed. The pending, queued, and active configurations are modified. This will immediately force an end-of-delay.
- If <stage> is NEXT, then a TIME:SCHedule will be performed. Both the pending and queued configurations are modified. The active configuration will be updated to the new queued configuration at the next end-of-delay.
- If <stage> is QUEUE, then only the pending configuration will be modified. An install or queue operation will need to be run for any changes to take effect.

Parameters:

- <stage> - May be ACTIVE, NEXT, or QUEUE to target the active, queued, or pending configurations respectively.
- <pulses> - The number of *additional* pulses that will be generated during a shot, as a 32-bit unsigned integer. All 32-bit values are valid. A value of 0 indicates that trains are disabled.

Response: OK or an error.

TRAI:n:<stage>:CouNT?

Description: Query the number of train pulses per trigger.

Parameters:

- <stage> - May be ACTIVE, NEXT, or QUEUE to target the active, queued, or pending configurations respectively.

Response: Reads the specified configuration (active, queued, or pending), then responds with the trains pulse count as an unsigned 32-bit unsigned integer. A value of 0 indicates that trains are disabled.

TRAIIn:<stage>:SPACE <time>

Description: Set train pulse spacing.

Sets the trains pulse spacing of the pending configuration.

Immediately after changing the pending configuration, a timing configuration install or queue may be performed, depending on the value of <stage>:

- If <stage> is ACTive, then a TIME:COMmit will be performed. The pending, queued, and active configurations are modified. This will immediately force an end-of-delay.
- If <stage> is NEXT, then a TIME:SCHedule will be performed. Both the pending and queued configurations are modified. The active configuration will be updated to the new queued configuration at the next end-of-delay.
- If <stage> is QUEue, then only the pending configuration will be modified. An install or queue operation will need to be run for any changes to take effect.

Parameters:

- <stage> - May be ACTive, NEXT, or QUEue to target the active, queued, or pending configurations respectively.
- <time> - The duration between repeated output pulses, as a time. Available suffixes are Ps (picoseconds), Ns (nanoseconds), Ms (milliseconds), and S (seconds). Default units are seconds. Scientific notation and SCPI-format numbers are accepted. Minimum increments are in picoseconds. Values 60 nanoseconds (inclusive) up to 85.899,345,940 seconds (exclusive) are allowed. This value is rounded down to the nearest multiple of 20 nanoseconds.

Response: OK or an error.

TRAIIn:<stage>:SPACE?

Description: Query train pulse spacing.

Parameters:

- <stage> - May be ACTive, NEXT, or QUEue to target the active, queued, or pending configurations respectively.

Response: Reads the specified configuration (active, queued, or pending), then responds with the configured train spacing, formatted as a time, in seconds. Time values are extended-precision floating values with units of seconds, which may require more than a double-precision floating point to represent full precision.

Frames Commands

TFRame:STATus?

Description: Report the frames engine status.

Response: Responds with the current running status of the frames engine, as a single word. Possible responses are:

- OFF - The frames engine is inactive.
- RUNNING - The frames engine is currently running.
- ERROR - The frames engine has been halted due to an internal error. Such a response is likely indicative of a hardware failure. In this state, the frames engine behaves like it is in a DONE state: active, but not running.
- DONE - The frames engine is active and has completed running.

TFRame:PoinTeR?

Description: Query the current frame pointer.

Reports the index of the currently-loaded frame in hardware. The next valid incoming trigger will use this frame's configuration to produce outputs.

Response: Responds with index the currently-active frame as a 16-bit unsigned integer.

Notes:

- The value of the frame pointer is undefined while the frames engine is inactive. Although the behavior matches the T560's, using this value while the frames engine is inactive is not recommended.

TFRame:REMAining?

Description: Query the number of frames remaining to be loaded.

Response: A decimal, 32-bit integer representing the number of frame loads remaining before the frame engine is in a DONE state. A value of 0xFFFFFFFF indicates that the engine is either currently in a DONE state or the remaining loop count represents infinity.

Notes:

- The value returned by this command should be considered undefined if the frames engine is inactive.

TFRame:LoadCouNT <n>

Description: Clear the loaded frame count.

Clears the loaded frame counter. This command may be run while the frames engine is active and running.

Parameters:

- <n> - May be any integral number. This number is ignored.

Response: OK or an error.

TFRame:LoadCouNT?

Description: Query the number of frames loaded into hardware.

Reports the number of frames loaded into hardware by the frames engine.

Response: The number of frames loaded while the frames engine has been active, since last power-on or TFRame:LoadCouNT <n> command execution.

TFRame:STArT

Description: Start frames operation.

Activates the frames engine, then starts it. If the frames configuration is valid, the frames engine will then be moved into the running state and the first frame will immediately be loaded into hardware.

The frames configuration is considered invalid if the index of the last frame to be executed (TFRame:LOOP:LAST?) is not greater than the first frame index (TFRame:LOOP:FIRST?). In this case, this command will produce a ?08: Settings conflict error and the frames engine will not be started.

Response: OK or an error.

Notes:

- This command terminates any possibly-running shot by issuing an end-of-delay (see: fEOD).
- To preserve T560 compatibility, running this command increments the loaded frame count by 2, instead of the expected 1.

TFRame:STOp

Description: Cancel frame mode.

Deactivate the frames engine and force an install.

Response: OK or an error.

Notes:

- This command issues an install as though the TIME:COMMit command were run, and carries all notes associated with that command.
- This command terminates any possibly-running shot by issuing an end-of-delay (see: fEOD).

TFRame:INSTALL <frame>

Description: Apply a frame's timing settings immediately.

Overwrites the pending configuration with the contents of the specified frame, (consisting of channel configuration, timing settings, and pulse trains configuration), then issues an install (functionally equivalent to issuing a INSTaLL command). Issuing this command is functionally equivalent to issuing a TFRame:QUEue <frame> command followed by a fEOD command.

Parameters:

- <frame> - The index of the frame to load configuration from.

Response: OK or an error.

Notes:

- This command modifies the pending and queued configurations.
- This command terminates any possibly-running shot.

TFRame:QUEue <frame>

Description: Queue a frame's timing settings to be installed at next EOD.

Overwrites the pending configuration with the contents of the specified frame, (consisting of channel configuration, timing settings, and pulse trains configuration), then issues a queue (functionally equivalent to issuing a TIME:COMMit command).

Parameters:

- <frame> - The index of the frame to load configuration from.

Response: OK or an error.

Notes:

- This command modifies the pending and queued configurations, and requires an EOD to take effect.

TFRame:STORE <frame>

Description: Store the pending configuration into the specified frame.
Saves the pending configuration to the specified frame.

Parameters:

- <frame> - The index of the frame to save to.

Response: OK or an error.

Notes:

- In order for this command to be operable, the frames engine must be inactive (a frames engine state of DONE does not count). If this condition is not met, this command will produce a ?07: Illegal operation error.

TFRame:LOOP:FIRST <frame>

Description: Set the start frame index.

Parameters:

- <frame> - The index of the first frame to be executed.

Response: OK or an error.

Notes:

- In order for this command to be operable, the frames engine must be inactive (a frames engine state of DONE does not count). If this condition is not met, this command will produce a ?07: Illegal operation error.

TFRame:LOOP:FIRST?

Description: Query the start frame index.

Response: The index of the first frame to be executed, as a 16-bit unsigned integer.

TFRame:LOOP:LAST <frame>

Description: Set the end frame index.

Parameters:

- <frame> - The index of the last frame to be executed.

Response: OK or an error.

Notes:

- In order for this command to be operable, the frames engine must be inactive (a frames engine state of DONE does not count). If this condition is not met, this command will produce a ?07: Illegal operation error.

TFRame:LOOP:LAST?

Description: Query the end frame index.

Response: The index of the last frame to be executed, as a 16-bit unsigned integer.

TFRame:LOOP:COUNT <count>

Description: Set the frames loop count.

Sets the number of loops the frames engine shall perform, with the starting and ending frames determined by the TFRame:LOOP:FIRST <frame> and TFRame:LOOP:LAST <frame> commands, respectively.

Parameters:

- <count> - The number of loops the frames engine shall perform, as a 16-bit unsigned integer. A value of 0 means that only one iteration will be performed, 1 means that two iterations will be performed, etc. A value of 65535 (or 0xFFFF) represents infinity, and the frames engine will never cease operation.

Response: OK or an error.

Notes:

- In order for this command to be operable, the frames engine must be inactive (a frames engine state of DONE does not count). If this condition is not met, this command will produce a ?07: Illegal operation error.

TFRame:LOOP:COUNT?

Description: Query the frames loop count.

Response: Responds with the number of loops the frames engine shall perform, as a 16-bit unsigned integer. A value of 0 means that only one iteration will be performed, 1 means that two iterations will be performed, etc. A value of 65535 (or 0xFFFF) represents infinity, and the frames engine will never cease operation.

TFRame:MEMory:CLEAr

Description: Clear the frames memory.

Write the default configuration to all frames.

Response: OK or an error.

Notes:

- In order for this command to be operable, the frames engine must be inactive (a frames engine state of DONE does not count). If this condition is not met, this command will produce a ?07: Illegal operation error.

Memory Commands

MEMory:LOAD[:NORMaL]

Description: Load the system configuration from nonvolatile memory.

Loading the stored configuration will ensure that all configuration is re-applied, ensuring that no prior commands have affected the unit's state. Exceptions to this are the oscillator configuration (VCXO trim), frames memory, network configuration, and serial configuration (serial baud rate). These exceptions are saved and loaded independently from the memory commands.

Completion of this command may result in immediate activity by the unit, since whatever configuration was saved is immediately applied.

Response: OK or an error.

Notes:

- This command modifies all configuration settings except active network and clock configuration.
- Frames memory is not affected by this command.
- Running this command stops frames engine operation, equivalently to running TFRame:STOp.
- This command terminates any possibly-running shot by issuing an end-of-delay (see: fEOD).

MEMory:LOAD:DEFault

Description: Load configuration defaults.

Loading the default configuration will ensure that all configuration is re-applied, ensuring that no prior commands have affected the unit's state. Exceptions to this are the oscillator configuration (VCXO trim), frames memory, network configuration, and serial configuration (serial baud rate). These exceptions are saved and loaded independently from the memory commands.

Completion of this command will result in the unit becoming idle due to the trigger source being set to remote.

Response: OK or an error.

Notes:

- This command modifies all configuration settings except active network and clock configuration.
- Frames memory is not affected by this command.
- Running this command stops frames engine operation, equivalently to running TFRame:STOp.
- This command terminates any possibly-running shot by issuing an end-of-delay (see: fEOD).

MEMory:LOAD:DEMO

Description: Load the demonstration configuration.

The demonstration configuration differs from the default configuration in that:

- The trigger source is set to internal (as opposed to remote)
- The predivider divisor is set to 4000 (as opposed to 0)

Completion of this command will result in the unit outputting pulses immediately due to this configuration.

Response: OK or an error.

Notes:

- This command modifies all configuration settings except active network and clock configuration.
- Frames memory is not affected by this command.
- Running this command stops frames engine operation, equivalently to running TFRame:STOp.
- This command terminates any possibly-running shot by issuing an end-of-delay (see: fEOD).

MEMory:SAVE[:ACTive]

Description: Save the current configuration to nonvolatile memory.

All settable configuration is saved to non-volatile memory immediately. Exceptions to this are the oscillator configuration (VCXO trim), frames memory, network configuration, and serial configuration (serial baud rate). These exceptions are saved and loaded independently from the memory commands.

Response: OK or an error.

Notes:

- Unlike other memory commands, running this command does not impact device operation.
- Frames memory is not saved to non-volatile memory.
- Clock configuration is saved by this command, however VCXO trim is not.

Network Commands

SYSTem:COMMunicate:SERial:BAUD <baud>

Description: Set the serial port baud rate.

Configures the baud rate of the RS232 serial port, then saves this baud rate to non-volatile memory. The configured baud rate will also be applied at device power-on until reconfigured.

Parameters:

- <baud> - The baud rate, as an integer. Valid values are 9600, 19200, 38400, 57600, 115200, 230400, or 460800.

Response: OK or an error.

Notes:

- Take caution when running this command over the serial interface: the baud rate changes during command execution. Because of this, the OK response by this command will be output in the newly-configured baud rate.
- Running this command with a <baud> parameter equal to the current baud rate will not interrupt serial communications.

SYSTem:COMMunicate:SERial:BAUD?

Description: Query the serial port baud rate.

Response: Responds with the current operating RS232 serial port baud rate, formatted as an integer. This will be one of: 9600, 19200, 38400, 57600, 115200, 230400, or 460800.

SYSTem:COMMunicate:SOCKeT:ADDReSS <address>

Description: Set the IPv4 address for the pending network configuration.

Sets the pending network configuration's IPv4 address.

Parameters:

- <address> - The IPv4 address, in the form X.X.X.X. 0.0.0.0 indicates that DHCP shall be used to obtain an IPv4 address.

Response: OK or an error.

Notes:

- Invalid IP addresses other than 0.0.0.0 (such as those not matching the set IPv4 subnet mask) may produce undefined behavior and are not supported.

SYSTem:COMMunicate:SOCKeT:ADDReSS?

Description: Query the IPv4 address set by the active network configuration.

Query the pending IPv4 address configuration. Note that since this represents the pending configuration, not the current network status, a valid response is 0.0.0.0, indicating that DHCP is to be configured. At power-up, this is 0.0.0.0.

Response: The most recently applied IPv4 address configuration, formatted as "X.X.X.X" (with quotes).

SYSTem:COMMunicate:SOCKeT:APPLY

Description: Apply and save the pending network configuration.

Applies the pending network configuration to the active network configuration. This will also immediately terminate any network connections to the device while the Ethernet connection is restarted. Once applied, the network configuration will be saved to non-volatile memory.

Response: OK or an error.

Notes:

- Running this command from network sessions will terminate the session from which the command was issued; there is no guarantee that a response will be sent back over the network interface (it likely will not).
- If the network configuration is set to DHCP using the `SYSTEM:COMMunicate:SOCKET:ADDRESS <address>` command, the value set by the `SYSTEM:COMMunicate:SOCKET:SubnetMASK <mask>` command is effectively ignored, and may not be saved to non-volatile memory.
- Running this command immediately after power-on will set the network configuration to DHCP, due to the power-on default values of the pending network configuration.

`SYSTEM:COMMunicate:SOCKET:HostNAME?`

Description: Query the device hostname.

Response: Responds with the device's serial number in the format `T660-XXXXX`, where `XXXXX` is the serial number. This returned string is the name with which the device may be accessed using DHCP and uDNS.

Notes:

- Unlike other networking queries, the response produced by this command does not include quotation marks (").

`SYSTEM:COMMunicate:SOCKET:MAC?`

Description: Query the device MAC address.

Response: Responds with the device's Ethernet MAC address in the form `"XX:XX:XX:XX:XX:XX"` (with quotes).

`SYSTEM:COMMunicate:SOCKET:NetStat?`

Description: Report the current network status.

Response: Responds with the current network status (not to be confused with the active network configuration) in the form `"<ip_address>","<subnet_mask>","<DHCP_status>","<link_status>"`, where:

- `<ip_address>` - The IPv4 address, formatted as `X.X.X.X`, or Pending if DHCP is enabled and the device does not yet have an IP address.
- `<subnet_mask>` - The IPv4 subnet mask, formatted as `X.X.X.X`, or Pending if DHCP is enabled and the device does not yet have an IP address.
- `<link_status>` - UP, if there is a physical link detected via the Ethernet port, or DOWN otherwise.
- `<DHCP_status>` - DHCP if the current configuration is DHCP, or STATIC otherwise.

Notes:

- An ethernet connection to an invalid network will still be reported with a link status of UP - this refers purely to the physical connection detection and does not reflect network connectivity.

SYSTem:COMMUnicate:SOCKeT:SubnetMASK <mask>

Description: Set the IPv4 subnet for the pending network configuration.

Parameters:

- <mask> - The IPv4 subnet mask, in the form X.X.X.X.

Response: OK or an error.

Notes:

- Invalid subnet masks (such as those not matching the set IPv4 address) may produce undefined behavior and are not supported.
- If the network configuration is set to DHCP using the SYSTem:COMMUnicate:SOCKeT:ADDReSS <address> command, this value is effectively ignored once applied, and may not be saved to non-volatile memory.

SYSTem:COMMUnicate:SOCKeT:SubnetMASK?

Description: Query the IPv4 subnet set by the active network configuration.

Response: The pending IPv4 subnet configuration, formatted as "X.X.X.X" (with quotes). At power-up, this is 255.255.255.0.

Notes:

- If the network configuration is set to DHCP as reported by the SYSTem:COMMUnicate:SOCKeT:ADDReSS <address> command, the reported value may be the default value.